

A Cost-Effective Predictive HR Analytics Framework for Sri Lankan SMEs Using Cloud-Native Serverless AI

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Abstract

Small and medium enterprises employ close to half of Sri Lanka's workforce but administer human resources manually, and the AI-driven platforms that could give them early warning of employee resignation are priced for enterprise budgets. This dissertation designs, builds and evaluates a cost-effective predictive HR analytics framework for firms of twenty to fifty employees, integrating a serverless attrition-prediction service into a production HR platform under an operational ceiling of LKR 10,000 per month.

Following Design Science Research, two models were trained and compared. A **transfer model** was fitted to 2,820 training records — 2,320 real international and 500 calibrated synthetic — and validated against held-out Sri Lankan data using only the four features the sources share. A **local model** was fitted within 230 Sri Lankan records on eight psychometric constructs. Both are served from an IAM-locked, scale-to-zero inference endpoint returning per-prediction SHAP explanations, and the constructs are captured inside the product through a weekly micro-survey, so that the stronger model has a route to its own inputs.

The central result is negative, and is reported as the principal finding. Attrition patterns do not meaningfully transfer across national contexts: the cross-context model reaches ROC-AUC 0.641 against the local model's 0.937, and decomposition shows that age and gender together score 0.457 — below chance — while the transfer model's residual skill comes almost entirely from satisfaction items that are, on the validation side, self-reported in the same instrument as the outcome. Synthetic augmentation contributes +0.032 ROC-AUC, within seed noise, and the sample-weighting scheme central to the data strategy proves immaterial at −0.007. Operational cost was measured rather than estimated at **LKR 4,050 per month**: inference is effectively free at 150 billable instance-seconds across four months, while the one component that cannot scale to zero, the database, accounts for roughly 70% of spend. Usability was not measured; the planned user study was replaced by expert inspection and the objective is reported as unmeasured rather than met.

Two limitations qualify every predictive claim: the Sri Lankan outcome is turnover *intention* rather than observed departure, and predictors and outcome derive from a single self-report instrument. The work contributes an empirical demonstration that international data and calibrated synthetic augmentation cannot substitute for local data collection in this domain, a

measured serverless cost baseline for SME-scale predictive analytics, and a deployed mechanism for capturing a psychometric model's inputs in production.

Keywords: employee attrition prediction, HR analytics, explainable AI, SHAP, serverless computing, cost-effective machine learning, transfer learning, turnover intention, small and medium enterprises, Sri Lanka

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Chapter 1 — Introduction

1.1 Background and Motivation

Small and medium enterprises form the backbone of Sri Lanka's economy, contributing approximately 52% of national gross domestic product and employing around 45% of the workforce [1]. Despite that weight, most Sri Lankan SMEs still administer human resources through manual, paper-based or spreadsheet-driven processes [2]. Under such reactive arrangements, the decisions that matter most — above all, identifying employees at risk of resignation — are taken only after the loss has already occurred, once recruitment, onboarding and knowledge-loss costs are unavoidable.

The global HR technology market answers this need with AI-driven platforms, but prices them out of reach. Per-employee-per-month licensing of roughly USD 8–15 amounts to LKR 240,000–450,000 annually for a fifty-employee firm [3] — a sum that excludes precisely the organisations with the least capacity to absorb turnover. Meanwhile, the economics of deploying machine learning have changed. Cloud services that scale to zero permit a pay-per-prediction model in which idle infrastructure costs nothing [4], creating an opening to deliver predictive analytics to resource-constrained organisations at a small fraction of conventional cost, consistent with Sri Lanka's national artificial intelligence and digital economy strategies [5], [6].

1.2 Problem Statement

Employee turnover falls disproportionately on smaller employers. Replacing a departing employee is estimated to cost between 50% and 200% of annual salary once recruitment, onboarding, training and lost productivity are accounted for [7]. In Sri Lanka this pressure is compounded by sustained skilled-worker emigration, with over 300,000 departures recorded annually [8], which makes retention of remaining staff a question of organisational survival rather than optimisation.

Machine learning has been shown to predict attrition effectively [9], [10], yet adoption among SMEs remains low, for three reasons. **Cost:** enterprise per-employee pricing assumes an enterprise budget. **Infrastructure:** self-hosted machine learning presumes hardware and operational expertise that a twenty-person firm does not have [11]. **Integration:** standalone

analytics tools create data silos disconnected from the systems where HR work actually happens [12].

A fourth barrier is specific to this context and became the central concern of this study. **There is no public, individual-level record of observed employee attrition for Sri Lanka.** Any model must therefore either borrow from other national contexts or be fitted to whatever local data exists — and whether such borrowing works is an empirical question that had not been tested.

The research gap is consequently a cost-optimised, cloud-native predictive analytics framework that is financially accessible, architecturally integrated into an existing HR system, and genuinely contextualised for the Sri Lankan workforce rather than merely deployed there.

1.3 Research Questions

- **RQ1** — What are the most significant predictors of employee attrition derivable for the Sri Lankan SME context, and to what extent do attrition patterns transfer from international data to Sri Lanka?
- **RQ2** — To what extent can a serverless cloud architecture reduce the operational cost of deploying HR predictive analytics relative to persistent infrastructure, and can it remain within a monthly budget of LKR 10,000?
- **RQ3** — Can a hybrid dataset approach, combining real international attrition data with calibrated synthetic data, produce a usable attrition classifier for the low-volume Sri Lankan SME data environment?

1.4 Aim and Objectives

Aim. To design, develop and evaluate a cost-effective, serverless, AI-driven predictive analytics module integrated into an existing cloud-native HR system, to forecast employee attrition risk for Sri Lankan SMEs.

- **O1 — Feature identification.** Identify predictors of attrition appropriate to the Sri Lankan SME context, in compliance with the Personal Data Protection Act No. 9 of 2022 [13].
- **O2 — Architecture design.** Design a serverless architecture maintaining low monthly operational cost for a twenty-to-fifty employee organisation, with a verifiable cost baseline against always-on deployment.

- **O3 — Model development.** Develop and train classification models using a hybrid dataset strategy, targeting a usable early-warning operating point with high recall on the attrition class.
- **O4 — System integration.** Implement a risk-insights interface within the HR product, including SHAP-based explanations.
- **O5 — Evaluation.** Evaluate predictive performance, deployment cost and usability.

1.5 Scope

In scope: binary attrition-risk classification; a serverless prediction service; an in-product risk-insights interface with per-prediction explanations; cost measurement; a calibrated synthetic-data pipeline; and privacy-preserving data handling.

Out of scope: real-time communications monitoring; mobile application development; modification of core payroll, leave or attendance logic; multilingual natural-language processing; and succession planning.

1.6 Contributions

This dissertation makes four contributions.

1. **An empirical demonstration that attrition models do not transfer across national contexts** on the demographic and satisfaction features that international and Sri Lankan data share. A locally fitted psychometric model reaches ROC-AUC 0.937 where a cross-context model reaches 0.641, and the gap survives correction for leakage, substitution of the estimator, and alternative binarisation of the outcome. Decomposition further shows that the cross-context model's residual skill is largely same-source correlation rather than transferred knowledge.
2. **A cost-viable serverless reference architecture**, measured rather than estimated, with the finding that inference is effectively free and the always-on database dominates operational cost.
3. **A working in-product data-capture mechanism** that generates the model's own input constructs, addressing the gap between a model that performs well in study conditions and one that could be operated.

4. **The reporting of negative and infeasible results** — the transfer failure, the immateriality of the sample-weighting scheme, and the structural impossibility of subgroup fairness validation on the available sample — rather than their omission.

1.7 Deviations from the Approved Proposal

Three departures from the approved proposal are declared here, before any result is presented, together with the reasoning behind each.

RQ1 shifted from behavioural to psychometric predictors. The proposal asked which *behavioural* predictors — attendance patterns, leave frequency, salary positioning, tenure, payroll anomalies — predict attrition in Sri Lankan SMEs. Answering that requires operational HR records paired with observed resignations, and no such Sri Lankan dataset exists publicly; nor could one be constructed within this project, since the deployed system has not yet accumulated real attrition outcomes. The only individual-level Sri Lankan instrument available measures psychometric constructs against turnover *intention* [14]. RQ1 was therefore reframed, first in the interim report, to ask which predictors are derivable for this context and how far international patterns transfer. The delivered local model consequently uses eight psychometric constructs rather than behavioural features. The behavioural-feature pipeline was nonetheless built and remains in the system, awaiting the outcome data that would make it trainable.

The RQ3 success criterion changed from $F1 \geq 80\%$ to a recall-primary operating point. The proposal committed to an F1 score of at least 80%. That criterion was replaced during the project by a high-recall target, on the reasoning set out in §3.8: in early warning, a missed resignation is far more costly than a false alarm, and F1 weights the two errors equally. The change is defensible, but it relaxed a stated numeric commitment and is declared as such. **Reported against the original criterion, RQ3 fails:** the hybrid model achieves an F1 of approximately 0.29 on Sri Lankan data, and the local model 0.699 at its honest operating point. Neither reaches 80%. Chapter 6 answers RQ3 negatively on both criteria.

The usability study was not conducted. Objective O5 included evaluation by the System Usability Scale. No participants were recruited and the timeline did not permit recruitment, so the usability target is **unmeasured — neither met nor refuted**. A heuristic expert inspection is

reported in its place (§5.10), together with an explicit statement of what that substitution can and cannot establish.

A fourth, smaller correction is noted for completeness: the interim report described age as dominating the transfer model. Subsequent analysis showed that age in the validation data takes only four coded values with 88.7% at a single one, so the apparent dominance is an artefact of coding rather than a substantive age effect (§5.3). That characterisation is withdrawn.

1.8 Structure of the Dissertation

Chapter 2 surveys the relevant literature on attrition prediction, explainable machine learning, serverless economics, and fairness in algorithmic employment decisions, and locates the gap this study addresses. Chapter 3 sets out the research methodology and the design decisions governing data, modelling and evaluation. Chapter 4 describes the implemented system. Chapter 5 reports testing and evaluation, including a five-part audit of the predictive claims and a measured cost study. Chapter 6 answers the research questions, states the contributions and limitations, and identifies future work.

Chapter 2 — Literature Review

This chapter surveys the five bodies of work this study draws on: machine learning for attrition prediction, explainable AI in employment decisions, the transferability of predictive models across contexts, the economics of serverless deployment, and the adoption of HR analytics in small firms and developing economies. It closes by locating the gap the study addresses. The review is organised to make one argument visible: the field has largely *assumed* that an attrition model learned in one organisational and national context carries to another, and has rarely tested that assumption directly. §2.3 develops this point, because the central empirical result of this dissertation is a test of exactly that assumption.

2.1 Machine Learning for Employee Attrition Prediction

Turnover research long predates machine learning. Mobley's withdrawal model [15] and the meta-analytic consolidation by Griffeth et al. [16] established the antecedents that still structure the field — satisfaction, commitment, perceived alternatives, tenure — and quantified their typically modest bivariate associations with leaving. Allen [7] translated this into practitioner terms, framing turnover as a cost centre amenable to management rather than an unavoidable overhead.

The machine learning contribution has been to treat attrition as a supervised classification problem rather than a hypothesis-testing one. Punnoose and Ajit [10] compared seven algorithms and reported that gradient-boosted trees substantially outperformed logistic regression, an ordering that has proved durable: tree ensembles remain the practical default for tabular HR data, for the reasons Breiman set out [17] and because class imbalance is tractable with established resampling methods [18]. Sarker [9] situates this within the broader applied-ML landscape.

The methodological difficulty is that this literature has consolidated around a single dataset. The IBM HR Analytics Employee Attrition dataset [19] — 1,470 records, 35 attributes — is the most widely used benchmark in the field, and recent work continues to rely on it exclusively. Li's transformer-based framework [20] and the explainable risk-scoring system of Pavithran and Vadivel [21] both train and evaluate on it alone. Two properties of that dataset deserve more attention than they usually receive: it is **fictional**, generated by IBM data scientists rather than observed, and it describes a single large US employer. A benchmark with those two properties

can establish that an algorithm works on data of a certain shape. It cannot establish that a model works on a real workforce, and still less on a workforce in a different economy.

2.2 Explainable AI in HR Decision-Making

Where a prediction affects a person's employment, the reasoning behind it becomes material rather than decorative. SHAP [22] provides a unified additive attribution framework grounded in Shapley values, and Lundberg et al. later gave an exact polynomial-time algorithm for tree ensembles [23], removing the approximation error that limits model-agnostic explainers. This matters for the present work: with a tree ensemble, per-prediction attributions are exact rather than sampled.

Applied HR systems have adopted this quickly. Mohiuddin et al. [24] combine SHAP attributions with what-if analysis in a decision-support system, so that a manager sees not only which factors drove a risk score but which changes would plausibly alter it — the step from explanation to intervention.

Rudin [25] supplies the necessary counterweight. Her argument is that post-hoc explanation of a black box is not equivalent to interpretability, and that in high-stakes domains an inherently interpretable model should be preferred outright, since an explanation that is merely plausible can legitimise a model that is wrong. Employment decisions fall squarely within her definition of high-stakes. The position adopted in this dissertation is a middle one, and is stated as such: a random forest with exact tree SHAP is neither a black box explained by a surrogate nor a genuinely interpretable model, and the explanations it produces are treated throughout as evidence about *the model's behaviour*, not as evidence about why an employee is actually likely to leave.

2.3 Cross-Context Transfer of Attrition Models

This section carries the study's central claim, so the argument is set out in three steps: what the ML literature implicitly assumes, what the statistical and organisational literatures predict, and what has actually been tested.

The assumption. Attrition studies are routinely written as though the model produced is a general instrument. A system trained and validated on one organisation's records is presented as

a method for predicting employee attrition, not as a method for predicting attrition *at that organisation*. The assumption is rarely defended because it is rarely stated. Where authors do acknowledge it, they defer it: Pavithran and Vadivel note that their dataset "might limit the model's capacity to generalize across different types of organizations" and recommend extension to "multi-organizational and industry-specific datasets" as future work [21], while Li's evaluation does not address cross-organisational generalisation at all [20]. The recurring call for validation on diverse geographic and sectoral data is itself evidence that such validation has not generally been performed.

What theory predicts. Moreno-Torres et al. [26] provide the vocabulary for what should be expected when a model is moved between populations, distinguishing covariate shift (the feature distribution changes), prior probability shift (the base rate changes), and concept shift (the relationship between features and outcome changes). Cross-national HR data plausibly exhibits all three: workforce age and tenure structures differ, attrition base rates differ, and — critically — the meaning of the predictors may differ. Concept shift is the damaging case, because it cannot be repaired by reweighting.

That concept shift is not hypothetical here is established outside the ML literature. Fischer and Mansell's cross-cultural meta-analysis [27] found that the relationships between forms of organisational commitment and turnover intention vary systematically with national cultural context — affective commitment relating more strongly to turnover intentions in individualistic settings, normative commitment more strongly in collectivistic ones. The organisational behaviour literature therefore predicts that the structural relationship an attrition model learns in one national context will not hold unchanged in another.

What has been tested. Very little. The ML literature assumes transfer and defers testing it; the OB literature predicts that transfer should partly fail but does not express this as a predictive-model evaluation. For Sri Lanka specifically, no published study evaluates an internationally-trained attrition model against local data. The gap is thus not an absence of models but an absence of an honest transfer test — and, as §1.2 noted, the deeper obstacle is that no public individual-level record of observed Sri Lankan attrition exists against which such a test could be run on behavioural outcomes.

2.4 Serverless Architecture and the Economics of Deployed ML

Machine learning as a service [11] removed the requirement that an organisation own infrastructure to use a model. Serverless computing extends this by removing the requirement to provision capacity at all: the Berkeley view [4] characterises the shift as an abstraction comparable to the move from assembly to high-level languages, with the operational consequence that idle capacity is not billed. Managed platforms package training and serving on the same basis [28].

The cost evidence is genuinely contested, and the disagreement is informative. Adzic and Chatley [29] document two industrial migrations in which moving to a function-based deployment reduced hosting costs substantially. Eivy and Weinman [30] argue the opposite tendency, showing that per-invocation pricing can grow rapidly even at modest traffic, that provider free quotas are insufficient for a service of any real size, and that the economics depend heavily on the execution profile of the workload.

Read together, these are not contradictory findings but a statement about workload shape. Serverless deployment is economically favourable where duty cycle is low and traffic is bursty, and unfavourable where load is sustained and predictable. Attrition inference for a fifty-employee firm — a handful of predictions per week, a monthly retrain — sits at the extreme favourable end of that spectrum, which makes it a strong candidate case rather than a general endorsement. Two qualifications follow and are carried into Chapter 5. First, scale-to-zero imposes cold-start latency, an engineering cost incurred in exchange for the economic one. Second, the argument applies only to components that *can* be made stateless; a relational database that must remain available cannot, and its cost is unaffected. Neither the general serverless cost literature nor the HR analytics literature has measured this trade-off for an SME-scale deployment in a low-income economy, where the absolute budget rather than the relative saving is the binding constraint [3].

2.5 HR Analytics in SMEs and Developing Economies

Angrave et al. [12] argued that HR was poorly positioned to exploit analytics, citing weak quantitative capability and tools designed around reporting rather than decision-making. Subsequent empirical work has borne this out and refined it for smaller firms. Ilyas et al. [31],

surveying 308 HR professionals in SMEs in a developing economy, found adoption driven by performance expectancy, effort expectancy, facilitating conditions and social influence, with perceived trust and supervisor support acting as significant additional determinants — that is, by perceived usability and organisational support rather than by predictive accuracy.

Bahanni et al. [32] interviewed 31 HR managers in Morocco and describe how external constraints — sociocultural norms and underdeveloped technological infrastructure — interact with internal weaknesses in data governance and inconsistent HR practice to produce compounding barriers. The Sri Lankan picture is consistent: Kirupananthan et al. [2] report low AI readiness among local SMEs, in an economy where SMEs contribute roughly half of GDP [1] and skilled emigration sustains turnover pressure [8].

The design implication is direct, and it shapes this project's contribution. For this population the binding constraints are cost, integration into existing workflow, and trust — not algorithmic sophistication. A more accurate model delivered as a separate tool at enterprise pricing does not address the barrier that the evidence identifies.

2.5.1 Evaluating Usability with Small or No Samples

Since usability rather than accuracy governs adoption in this setting, its measurement matters. The System Usability Scale [33] is the standard instrument, and Bangor et al. [34] supply the interpretive bands that make a raw SUS score meaningful by attaching adjective ratings to score ranges. Both require participants. Where participants cannot be recruited, Nielsen and Molich's heuristic evaluation [35] offers an expert-inspection alternative in which a small number of evaluators assess an interface against a fixed set of usability principles. The two methods are not equivalent — heuristic evaluation identifies plausible problems, whereas SUS measures experienced satisfaction — and Chapter 5 states explicitly what the substitution made in this study can and cannot establish.

2.6 Turnover Intention and Psychometric Constructs

The Sri Lankan validation data used in this study measures turnover *intention* rather than observed departure, so the standing of intention as a construct requires justification rather than apology.

Mobley [15] modelled withdrawal as a sequence — dissatisfaction, thoughts of quitting, evaluation of alternatives, intention to search, intention to quit, departure — in which intention is the proximal antecedent of the behaviour. Ajzen's theory of planned behaviour [36] generalises this: intention is the immediate determinant of volitional action. The empirical warrant is provided by Steel and Ovalle's meta-analysis [37], which reports a weighted average correlation of approximately **0.50** between behavioural intentions and actual turnover, and finds intention to be a better predictor of leaving than either job satisfaction or organisational commitment.

That figure is the honest basis for the proxy and also its limit. An association of 0.50 is the strongest available among the standard antecedents, which makes intention a defensible target; it also implies roughly a quarter of shared variance, which means intention is emphatically not a substitute for the observed outcome. Every claim in this dissertation about the local model is therefore a claim about intention, and is worded that way.

The specific instrument used here is that of Kanchana and Jayathilaka [14], who surveyed professionals in Sri Lankan startups across eight psychometric constructs — job satisfaction, work–life balance, happiness, management support, career management, innovative work behaviour, leader–member exchange and coworker support — with turnover intention as the outcome. Because predictors and outcome are captured in a single self-report instrument, common method variance is a structural feature of the design rather than an incidental risk; Podsakoff et al. [38] set out both the mechanism and the available remedies, and §5.4 reports what could and could not be ruled out here.

2.7 Fairness and Ethics in Algorithmic HR

Barocas and Selbst [39] established the foundational point that discriminatory outcomes in data-driven decision systems arise from the data and the target definition rather than from discriminatory intent, and that existing anti-discrimination doctrine maps awkwardly onto them. Raghavan et al. [40] examined 18 vendors of algorithmic pre-employment assessments and found bias-mitigation claims that were inconsistently specified and rarely independently verifiable — evidence that the gap between stated and demonstrated fairness in commercial HR AI is wide.

The formal results constrain what any such system can promise. Kleinberg et al. [41] and Chouldechova [42] independently proved that calibration and equalised error rates across groups cannot generally be satisfied simultaneously when base rates differ. A deployed system must therefore choose which fairness criterion it satisfies and disclose the choice; claiming to be fair without qualification is not available.

For this project the operative requirements are the four-fifths rule as a disparate-impact screen, and Sri Lanka's Personal Data Protection Act No. 9 of 2022 [13], which governs the lawful basis, minimisation and security of the personal data involved. §5.8 reports the fairness audit and, importantly, the point at which the available sample was too small to support one.

2.8 Synthesis and Research Gap

Four gaps emerge, and together they define the study.

Transfer is assumed, not tested. The attrition literature consolidates on a single fictional US dataset and defers cross-context validation to future work, while the cross-cultural organisational literature predicts that the learned relationships should not hold unchanged. No study evaluates an internationally-trained attrition model against Sri Lankan data.

Serverless economics are argued, not measured, at SME scale. The cost literature disagrees in a way that turns on workload shape, and low-duty-cycle HR inference is exactly the case not yet measured — least of all against an absolute budget in a low-income economy.

Adoption research and modelling research do not meet. Studies of SME adoption identify cost, integration and trust as the barriers; modelling studies optimise accuracy on a benchmark. Little work delivers a model inside the system where HR work actually happens, at a cost the target firms could pay.

The data required does not exist. No public individual-level record of observed Sri Lankan attrition is available, which is why the alternatives this study evaluates — international transfer and calibrated synthetic augmentation — are the ones a resource-constrained developer would actually reach for, and why establishing whether they work is worth doing.

Chapter 3 sets out the methodology through which these gaps are addressed.

Chapter 3 — Methodology and Design

3.1 Research Approach

This project is conducted as Design Science Research (DSR), the paradigm appropriate to enquiry in which the construction and evaluation of an information technology artifact is itself the contribution to knowledge [43]. The artifact is a cost-effective predictive HR analytics framework for Sri Lankan small and medium enterprises: an operational HR platform, a machine-learning pipeline, and a serverless inference service, integrated and deployed. DSR is preferred to a purely experimental design because the research question is not only *whether* attrition can be predicted, but whether it can be predicted **within an economic and infrastructural envelope that a Sri Lankan SME could actually sustain**. That second condition cannot be evaluated without building the thing.

The work is structured according to the Design Science Research Methodology (DSRM) process model [44], whose six activities map onto the project as shown in Table 3.1. The mapping is given explicitly because DSR is frequently invoked as a label rather than followed as a process, and because it makes the evaluation obligations of each phase auditable.

Table 3.1 — DSRM process model [44] mapped to project phases

DSRM activity	Instantiation in this project	Artifact produced	Evaluation
1. Problem identification and motivation	SME attrition cost in Sri Lanka; absence of affordable predictive HR tooling; no local attrition dataset	Problem statement, research questions (Ch1)	Literature survey (Ch2)
2. Define objectives of a solution	Three measurable targets: recall > 80%, usability, operational cost < LKR 10,000/month	Objective specification (Ch1 §1.4)	Objectives restated as testable metrics (Ch5)
3. Design and	Multi-source data	Datasets, models,	Internal validation

development	strategy; two-model design; HR platform; serverless inference; Pulse Check capture	deployed system (Ch4)	during development
4. Demonstration	Deployment to Google Cloud Run with the HR application consuming live predictions	Running integrated system (Ch4 §4.5–4.7)	Functional verification
5. Evaluation	Held-out validation; five-part evaluation audit; cost attribution study	Reproducible reports under reports/ (Ch5)	Ch5 in full
6. Communication	This dissertation and the accompanying defence	Dissertation, source repository	Examination

3.2 Research Design: A Two-Model Comparison

The central design decision is that **two models are built and evaluated side by side rather than one**. This follows directly from the data situation described in §3.3: no public, individual-level record of observed employee attrition exists for Sri Lanka, while substantial real datasets exist for other national contexts.

The first model, referred to throughout as the **transfer model**, is trained on the pooled international training data and validated against the held-out Sri Lankan sample. It is necessarily restricted to the four features the two bodies of data share — age, gender, job satisfaction and work-life balance. It tests the proposition that a model learned in one labour market can be applied in another.

The second, the **local model**, is trained and cross-validated entirely *within* the Sri Lankan sample, using the eight psychometric constructs available there. It tests what is achievable when the data is local but small.

The contrast between the two is the empirical contribution of this work, and the design deliberately admits the possibility of a negative result. A single-model design that reported only

the stronger figure would conceal the more interesting finding; presenting both makes the limits of cross-context transfer visible and measurable.

One asymmetry must be stated at the outset rather than discovered by the reader in Chapter 5. The international training data records **observed attrition** — the employee left. The Sri Lankan validation data records **turnover intention**, a self-reported psychological state [14]. Intention is a well-established antecedent of turnover but is not turnover [16]. The two models therefore do not predict identically defined outcomes, and this label shift is a confound on the transfer comparison that is carried explicitly through Chapters 5 and 6 rather than resolved.

3.3 Data Strategy and Source Selection

No public individual-level HR attrition dataset exists for Sri Lanka. The closest available instrument is a 2023 survey of 230 Sri Lankan startup professionals measuring turnover intention on a five-point ordinal scale [14]. A model trained exclusively on Sri Lankan data is therefore not possible.

The strategy adopted is **multi-source weighted training**: train on real international data drawn from developing-economy and private-sector contexts, supplement with a smaller volume of synthetic records calibrated to Sri Lankan conditions, and hold the Sri Lankan survey back entirely for validation.

Table 3.2 — Multi-source data strategy

Source	Records	Type	Weight	Role
Saudi employee attrition	1,191	Real, developing-country private sector	2.0	Training
Russian employee turnover	1,129	Real, private-sector company	2.0	Training
Calibrated synthetic (SL context)	500	Simulated, Sri Lankan SME	0.5	Training
Sri Lanka startup	230	Real, turnover	—	Held-out validation

survey [14]		intention		
IBM HR Analytics [19]	1,470	Synthetic benchmark	—	Published comparison only

The training master set therefore contains 2,820 records. Real records carry a sample weight of 2.0 and synthetic records 0.5, so that observed human behaviour dominates the fitted decision boundary and simulated data can inform it without governing it. The weights are passed to the estimator's `sample_weight` parameter rather than being applied by duplication, which preserves the true sample size for cross-validation.

Two exclusions are deliberate. The widely used IBM dataset [19] is **not** used for training. It is a synthetic dataset published for demonstration purposes, and including it would inflate apparent performance while adding no real-world signal; it is retained solely so that results can be compared against the large published literature that uses it. Second, the Sri Lankan sample is treated as a **held-out validation set rather than being pooled and cross-validated**, so that no Sri Lankan record enters model fitting directly. One qualification on that statement is owed and is given in §3.5: the sample did contribute to the calibration step that shapes the synthetic component of the training data.

The resulting composition is set out in Table 3.2.

3.4 Normalisation Decisions

Three normalisation decisions govern cross-source comparability. Each is recorded here because each would silently corrupt the model if made differently, and because they are the decisions most likely to be probed in examination.

Income is z-score normalised within each source before merging. The sources report salary in different currencies and different economic periods; a raw merge would teach the model that Russian salaries are numerically larger than Saudi ones, which is an artefact of denomination rather than a fact about employees. Normalising within source means the model learns an employee's *relative salary position within their own labour market*, which is the construct actually of interest.

Ordinal features are min-max rescaled within each dataset. Satisfaction and work-life balance are recorded on a one-to-three scale in the training sources but a one-to-five scale in the Sri Lankan survey. Rescaling each to a common interval within its own dataset makes the transfer comparison meaningful. The rescaling bounds are persisted alongside the trained model so that inference applies exactly the transformation used in training — without this, train/serve skew would appear as unexplained degradation in production.

Missing values are encoded as `NaN`, never as zero. The sources overlap only partially on features, so absence is common and structural. Encoding a missing satisfaction score as zero would assert *minimum satisfaction* rather than *unknown satisfaction*, injecting a false and systematically biased signal. Imputation is instead performed inside the cross-validation fold (§3.8) so that no information crosses the train/test boundary.

3.5 Calibration and Synthetic Generation

Synthetic records are not drawn from arbitrary distributions. A logistic regression is first fitted to the pooled real records to estimate the direction and magnitude of each available predictor's association with attrition; the resulting coefficients are persisted to a calibration file and used to drive generation, so the synthetic sample reproduces relationships observed in real data rather than assumed ones.

A leakage disclosure is required here. The calibration was fitted to **2,550 records, and those 2,550 include the 230 Sri Lankan validation records** alongside the 2,320 international ones — the pipeline passes all three sources to the calibration step. There is therefore an indirect path by which the validation sample influenced the training data: validation records shaped the calibration coefficients, the coefficients generated the 500 synthetic records, and those records entered the training master. No Sri Lankan record was seen by either model during fitting, but the strict claim that the validation set is untouched by training does not hold, and it is stated here rather than left for a reader to derive.

The exposure is bounded and its direction is known. The leak is aggregate-level — eight regression coefficients estimated across 2,550 records — and it reaches the models only through the synthetic component, whose total contribution §5.5 measures at +0.032 ROC-AUC, inside seed noise. Any effect can only have *helped* the transfer arm, which is the arm this dissertation

reports as failing. The headline negative result in §5.3 is therefore conservative with respect to this defect, not flattered by it. A corrected pipeline would fit calibration on the international sources alone, and that is recorded in §6.4.

The intercept is recomputed rather than inherited, so that the generated sample carries a prevalence appropriate to a Sri Lankan SME context instead of the base rate of the source populations. Where a source dataset is unavailable, the pipeline degrades to literature-derived default coefficients rather than failing, which keeps the pipeline reproducible on a fresh machine.

The honest limitation is that synthetic data of this kind can only encode what the calibration captured. This is not left as an assertion: an ablation reported in Chapter 5 finds that a model trained on synthetic data *alone* performs at approximately chance, and that the synthetic augmentation contributes a small, non-harmful improvement rather than a substantive one. The synthetic component is therefore presented as augmentation, never as a substitute for real local data.

3.6 Construct Computation and Feature Sets

The Sri Lankan instrument measures eight psychometric constructs — job satisfaction, work-life balance, happiness, management support, career management, innovative work behaviour, leader-member exchange and coworker support. Each construct is computed as the **mean of its constituent survey items**, a definition applied identically in the offline preprocessing pipeline and in the production Pulse Check capture, so that a construct value means the same thing at training time and at inference time.

The dependent variable requires its own decision. Turnover intention is captured by four items; these are averaged into a composite and the composite is thresholded at **3.5 on the five-point scale** to produce the binary label, yielding a positive class prevalence of 14.3%. Any threshold on a continuous construct is a modelling choice rather than a natural boundary, and Chapter 5 reports a sensitivity analysis across alternative cut-points to establish that the study's conclusions do not depend on this one.

Two feature sets follow from §3.2: four shared demographic and satisfaction features for the transfer model, and the eight constructs for the local model. The eight constructs are **survey-sourced**, not operational HR data — a point with direct architectural consequences, since it

means the production system must capture employee self-report to feed the local model at all. That requirement is what the Pulse Check component (Chapter 4) exists to satisfy.

3.7 Model Selection and Training Protocol

The estimator is a Random Forest [17], configured with 400 trees and balanced class weighting. Random Forest was selected over managed automated machine-learning services for three reasons: it admits exact rather than approximated explanation; it imposes no per-prediction service cost, which is material to the cost objective; and it is inspectable, which matters for an employment-related application where an unexplainable decision is an ethical as well as a technical liability.

Class imbalance is addressed by SMOTE-family resampling combining synthetic minority over-sampling [18] with Tomek-link cleaning of the resulting boundary [45]. Both a resampling path and a class-weighting path are retained and reported, since the two are not equivalent and the choice between them affects the operating point.

Explanations are generated with SHAP TreeExplainer [22], which computes exact Shapley values for tree ensembles rather than sampling approximations. Explanations are produced per prediction, not only in aggregate, because the intended use is a manager asking why *this* employee has been flagged.

The selection of Random Forest is treated as a claim requiring evidence rather than a preference. Chapter 5 reports a controlled comparison against logistic regression and gradient boosting on identical data, folds and seeds.

3.8 Evaluation Design

Evaluation follows repeated stratified five-fold cross-validation, averaged across five random seeds, with the seed set fixed and recorded so that every reported figure is reproducible.

Three controls guard against optimistic bias. First, **imputation and scaling are performed inside each fold** by means of a pipeline, so that no statistic computed from held-out data can influence the model that is evaluated on it. Second, **decision thresholds are selected nested within the cross-validation** rather than tuned on the same data used to report performance; a threshold chosen on the reported split flatters precision and recall simultaneously, and Chapter 5

quantifies exactly how much. Third, **confidence intervals are estimated by bootstrap resampling** [46] rather than reported as point estimates, so that the reader can distinguish a real difference from sampling noise on a 230-record sample.

The primary metric is **recall on the attrition class**, because the application is early warning and the costly error is the employee who leaves unflagged. Recall is never reported alone: precision, F1, ROC-AUC, PR-AUC and confusion matrices accompany it, since a trivial classifier that flags everyone achieves perfect recall.

A reproducibility convention governs the evaluation: **each evaluation question is answered by one dedicated script that writes a machine-readable report**. No figure enters this dissertation that cannot be traced to a script and regenerated.

3.9 Ethical, Legal and Privacy Design

This project collected no primary data from human participants. All model training and validation data are secondary, previously published datasets, used under the ethics approvals obtained by their originating studies and cited in §3.3. The Pulse Check instrument described in Chapter 4 was implemented and functionally verified using synthetic test data only; no employee responses were collected. The ethical approval requirement in the KIU project guidelines is therefore not triggered.

Privacy-by-design measures nonetheless govern the architecture, in anticipation of real deployment and in compliance with the Sri Lanka Personal Data Protection Act No. 9 of 2022 [13]. Personally identifying information — names, national identity card numbers, email addresses, telephone numbers, bank details — is stripped **at the database layer**, before any data leaves the HR application; the export produces only modelling features with hashed identifiers. Access to export functions is restricted by role, every data operation carries a correlation identifier for audit, and data is encrypted in transit and at rest. Automated pre-transfer inspection for residual identifiers is specified in the design as a control on cross-border movement under PDPA §25; it is **specified but not yet provisioned**, and is recorded as such in the limitations rather than claimed as implemented.

One design decision is ethical rather than technical and is worth stating explicitly: **employees do not see their own risk score**. The Pulse Check returns only a confirmation of submission. A

system that reports a personal attrition-risk score back to the individual invites both distress and gaming, and neither serves the intervention purpose the tool exists for.

Protected attributes receive a decision rather than a default. Age and gender are inputs to the transfer model and are excluded from the local model. Chapter 5 reports a fairness audit that tests whether they can be removed without cost, and whether they can be reconstructed from the remaining features.

3.10 Cost Evaluation Methodology

The cost objective requires its own method, because the obvious approach does not work. The deployment shares a Google Cloud project with four unrelated systems, and a single database instance serves six databases of which two belong to this project. A project-level billing export would therefore charge this research for infrastructure it does not use, by an unknown and unauditable margin. Resource-level attribution from billing records would require the detailed billing export, which is not enabled on the account and cannot be applied retrospectively.

Cost is therefore established by **measuring usage per resource and applying published unit prices**. Per-service utilisation is retrieved from cloud monitoring telemetry over a four-month window and multiplied by the published rate for each service, with results reported under two scenarios: the project's attributed share of the shared deployment, and a dedicated single-tenant deployment of the kind an SME would actually operate. Each is reported both with and without the provider's free-tier allowance.

This is argued as the more faithful method rather than a substitute for billing data. The research question concerns what a Sri Lankan SME would pay to run this system, and such an organisation would deploy it alone; per-resource attribution models that deployment, whereas a shared invoice does not. The limitations — that unit prices are published rates rather than invoiced amounts, that shared-resource apportionment involves an assumption, and that measured traffic is development-scale rather than a live workload — are reported alongside the result in Chapter 5.

Chapter 4 — Implementation

4.1 System Architecture

The framework is implemented as **two systems joined across an API boundary** rather than as a single application. The first is an operational HR platform that a small enterprise uses daily for payroll, leave and attendance. The second is a machine-learning pipeline and a serverless inference service that consumes anonymised features and returns an attrition-risk assessment with explanations.

The separation is deliberate and is the central architectural decision of the implementation. An HR platform is a stateful, transactional, always-available system; an inference service is stateless, bursty and idle most of the time. Merging them would force the inference workload to inherit the availability profile — and therefore the cost profile — of the transactional system, which would defeat the economic objective. Keeping them apart allows the inference tier to scale to zero when nobody is asking it anything, which §4.5 shows is where the cost saving actually comes from.

The two systems also live in **separate repositories with no shared version history**. The HR platform is developed independently of this research; the machine-learning work integrates with it over HTTP rather than by merging codebases. This is a constraint that turned out to be a benefit: it forced the integration to be defined by a contract rather than by shared internals, which is the arrangement an SME adopting the framework would actually have.

4.2 The HR Platform

The HR platform is a TypeScript monorepo: a Node.js and Express REST API backed by PostgreSQL through the Prisma ORM, and a React single-page application built with Vite. The data model spans more than twenty entities across twenty-one versioned migrations, covering companies, users, employees, attendance, leave requests and balances, payslips, bank-file exports, rosters, shift templates, employee documents and audit logs.

Access control is layered. Authentication issues short-lived access tokens with rotating refresh tokens, and additionally supports passwordless biometric login via WebAuthn. Authorisation is role-based across four roles — owner, administrator, HR staff and employee — with the last

restricted to self-service. Every request passes through the same lifecycle: schema validation, authentication, role authorisation, controller, service, database.

One rule governs the entire data layer: **every database query filters by the authenticated user's company identifier**. Multi-tenancy is enforced explicitly in each service method rather than implicitly by middleware, on the principle that an isolation rule which is visible at every call site is harder to omit than one applied invisibly somewhere upstream. For a system holding salary and personal data for multiple companies, a cross-tenant leak is the most serious failure available, and the design accepts verbosity as the price of making that failure obvious.

The frontend lazy-loads every route and splits vendor bundles, so an SME on a constrained connection downloads only the screens actually used. Server state is managed through a query cache with revalidation rather than hand-rolled fetching, and dashboard aggregates are served from an in-memory cache keyed by company identifier, with the cache interface written so that it can be replaced by a shared cache without touching call sites. Figure 4.1 shows the resulting dashboard.

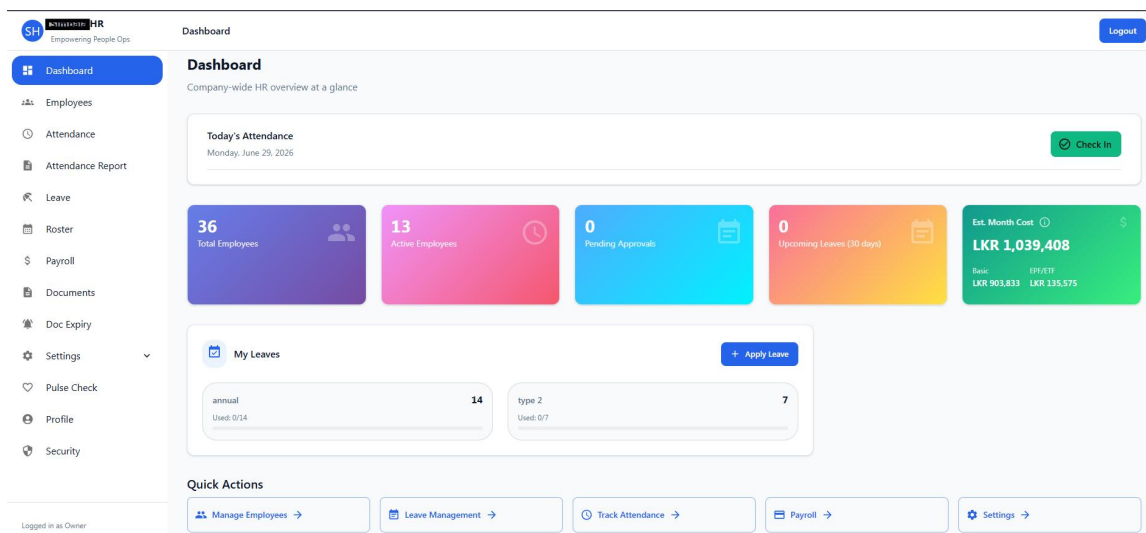


Figure 4.1 — The HR platform dashboard, showing headcount, leave utilisation and attendance aggregates.

4.3 Sri Lankan Statutory Compliance

The platform implements Sri Lankan payroll law directly rather than through a configurable generic engine, because the statutory rates are stable and an SME needs them correct by default rather than correct after configuration. Employees' Provident Fund contributions are calculated at

8% from the employee and 12% from the employer, the Employees' Trust Fund at 3%, all on basic salary, with Pay As You Earn applied on the progressive schedule published by the Inland Revenue Department.

Statutory leave minimums are enforced as floors that a company's configuration cannot fall below: fourteen days annual leave, seven days casual and seven days medical, the last accruing against the employment anniversary. Payslips are generated as PDFs, and salary disbursement files are exported in the CIPS and SLIPS formats used by Sri Lankan banks. Every payroll and bank-file operation writes an audit record.

This compliance layer is not incidental to the research. A predictive tool that an SME must run *alongside* its payroll system is a tool that will not be adopted; the prediction has value precisely because it sits inside the system where the HR work already happens.

4.4 The Machine-Learning Pipeline

The offline pipeline is implemented as five Python scripts executed in sequence, each with a single responsibility and a file-based contract with the next. Source datasets are retrieved and staged; raw survey spreadsheets are converted into clean numeric form with the eight psychometric constructs computed as means of their constituent items; a logistic regression is fitted to the pooled real records to produce calibration coefficients; synthetic records are generated from those coefficients; and finally the sources are merged into the weighted master training set, the held-out validation set and the benchmark comparison set.

Two engineering decisions in this pipeline are worth recording. First, the merge step **asserts target integrity before writing anything** — that the label contains no missing values, that it is strictly binary, and that every contributing source has supplied labelled rows. The three sources encode their outcome differently, one of them embedding it in an event field and another carrying leading whitespace in its categorical values, so the raw outcome columns are stripped and a single canonical binary label is derived. Silent corruption of the target is the most damaging error available in a supervised pipeline, and the assertions exist to make it loud.

Second, the pipeline **degrades rather than fails** when a source dataset is unavailable, substituting literature-derived default coefficients. This keeps the work reproducible on a

machine that cannot reach every upstream source, which matters for examination and for anyone reproducing the results later.

Training itself produces both models described in §3.2, together with a machine-readable training report and SHAP summary plots. Model artifacts are serialised with their fitted preprocessing state — imputation statistics and the per-dataset rescaling bounds — so that inference can reproduce training-time transformation exactly.

4.5 Serverless Inference

The inference service is a FastAPI application deployed to Google Cloud Run, serving both models on separate routes. Each prediction returns the probability, the operating threshold, the resulting flag and risk band, and the per-request SHAP contributions that explain it. Model bundles are loaded from Cloud Storage at start-up, with a copy baked into the container image as a fallback so the service remains available if object storage is briefly unreachable.

Four properties of the deployment matter to the research argument:

- **It scales to zero.** No instance runs when no prediction is being requested. §5.11 shows the inference service consumed 150 billable instance-seconds across four months, which is the mechanism by which the cost objective is met.
- **It is closed by default.** The service rejects unauthenticated invocation; the HR backend calls it with an identity token minted from the instance metadata service. The service account it runs under holds read access to one storage bucket and nothing else.
- **Retraining is scheduled, not manual.** A Cloud Run job re-runs training monthly under a scheduler trigger, so the model does not silently age.
- **Train/serve parity is enforced by construction.** The serving code applies the same gender encoding, the same persisted rescaling bounds and the same imputation as training, and the serving environment pins the identical scikit-learn version, since a version mismatch can break deserialisation of the model bundle outright.

The cold-start behaviour required an explicit decision. A scaled-to-zero service makes the first request after an idle period wait for a container to start. The obvious remedy — keeping one instance permanently warm — would have removed the delay and simultaneously destroyed the cost argument, since a warm instance bills continuously. The implementation instead uses an

extended request timeout with a single automatic retry: the first attempt wakes the instance, the retry arrives to a warm one. This is an engineering decision made in service of a research claim, and it is the clearest illustration in the implementation of the trade-off the framework is built around.

The deployment environment is a shared development project rather than a dedicated production tenancy. This has no effect on functional behaviour but has a direct effect on cost measurement, and is the reason §3.10 specifies per-resource attribution rather than project-level billing.

4.6 Integration with the HR Application

Integration was delivered as a thin proxy in the HR backend. The proxy exposes attrition endpoints under the application's own API, validates request payloads against a schema, and gates access behind a dedicated permission granted to owners automatically and to administrators explicitly. It obtains an identity token for the inference service at call time and forwards the request. When the inference service address is not configured, the feature disables itself cleanly rather than erroring — so the HR platform remains fully functional for a deployment that has not adopted the predictive component at all.

On the frontend, an attrition risk card appears on the employee detail page, presenting the eight construct inputs, the resulting probability against its threshold, the risk band, and the strongest contributing factors from the SHAP output. The card carries a visible caveat stating that the inputs are survey-sourced and that the model predicts turnover *intention* rather than departure. Figure 4.2 shows the card after a prediction has been requested.

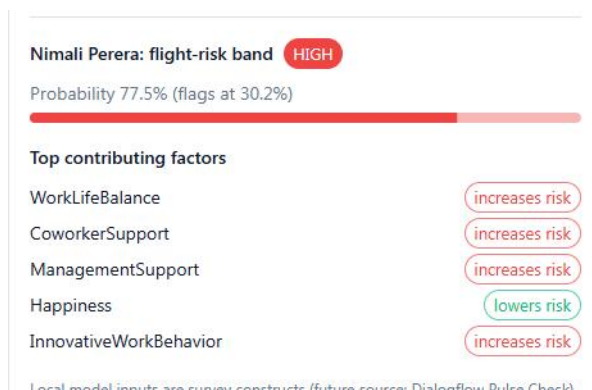


Figure 4.2 — Attrition risk card after a prediction, showing the risk band, the probability against its disclosed threshold, and the leading SHAP contributions.

The integration was verified end to end against the deployed services with a real authenticated session: the status endpoint reporting the feature enabled, both prediction routes returning coherent probabilities with corresponding SHAP attributions, and malformed input rejected by schema validation.

4.7 The Pulse Check

The local model consumes eight psychometric constructs, and those constructs do not exist in operational HR data. Attendance records and payroll history cannot tell you whether an employee feels supported by their manager. Without a mechanism to capture self-report, the stronger of the two models would be undeployable — an accurate predictor with no way to obtain its inputs.

The Pulse Check is that mechanism: a weekly sixteen-item Likert micro-survey, two items per construct, averaged into the eight constructs using **the same mean-of-items definition applied during training**, which is what preserves train/serve parity. One submission is stored per employee per ISO week, holding the raw answers, the derived constructs and a cached prediction. Scoring is best-effort by design — if the inference service is cold or unavailable, the response is still recorded and the submission never fails.

A full conversational agent was considered for this role and rejected. It would have added a managed service dependency and a recurring cost to capture data that a form in the existing application captures adequately, which would have undermined the cost objective the framework exists to demonstrate. The survey as presented to employees is shown in Figure 4.3.

Weekly Pulse Check
A 2-minute check-in on how work feels this week

✓ You've completed this week's Pulse Check. Thank you! You can update your answers below if anything has changed.

📘 Your honest answers help us understand and improve the employee experience. Tell us how much you agree with each statement (1 = strongly disagree, 5 = strongly agree).

1. I feel satisfied with my current job.

Disagree Neutral Agree

2. My work gives me a sense of accomplishment.

Figure 4.3 — The weekly Pulse Check survey presented to employees.

Two design decisions carry ethical weight. First, **employees never see their own risk score**; the survey returns only a confirmation of submission. A tool that reports a personal attrition-risk figure back to the individual invites distress and invites gaming, and serves no intervention purpose. Second, the manager-facing readout, shown in Figure 4.4, is gated behind the same permission as the attrition feature, so pulse-derived risk is not visible to peers.



Figure 4.4 — Manager-side pulse-derived risk readout on the employee detail page.

The evidential status of this component must be stated precisely. The Pulse Check was implemented, deployed and functionally verified end to end — question delivery, submission, construct derivation, scoring, persistence and manager readout — but it was **never populated with responses from real employees**. All data exercised through it is synthetic test data. Under a design science framing this remains a legitimate artifact contribution: the mechanism exists, works, and closes the loop between the research model and a deployable system. What it does not constitute is empirical evidence about real employee sentiment, and no claim of that kind is made anywhere in this dissertation.

Two further limitations follow from the same fact. The two-items-per-construct short form is not the full validated battery from which the constructs originate, and is therefore a noisier estimate; and because no responses were collected, the short form's internal consistency could not be assessed. Both are disclosed in the application's own interface as well as here, and both are carried into Chapter 6 as future work.

Chapter 5 — Testing and Evaluation

5.1 Evaluation Protocol

Evaluation proceeds on two levels. The software artifact is verified by automated testing (§5.2); the predictive claims are evaluated against the held-out Sri Lankan sample of 230 records containing 33 positive cases, a prevalence of 14.3% (§5.3–§5.8). The objectives carrying measurable targets — O3, O5 and O2 — are then assessed individually (§5.9–§5.11) and the whole is qualified by an explicit statement of validity threats (§5.12).

The remaining two objectives are design objectives rather than measurable ones, and are evidenced rather than scored. **O1**, feature identification under the Personal Data Protection Act, is discharged by the feature selection in §3.6 and the privacy architecture in §3.9, with the one control specified but not provisioned recorded in §6.3. **O4**, integration of a risk-insights interface into the HR product, is evidenced by the deployed and end-to-end verified components described in §4.6 and §4.7, and is qualified by the interface defects that §5.10 goes on to report. Neither carries a numeric criterion, and neither is claimed to have one.

Every quantitative result in this chapter is produced by a named script that writes a machine-readable report, and no figure appears here that cannot be regenerated by running it. This convention is what allows the limitations reported below to be stated with confidence rather than hedged.

5.2 Software Testing and Quality Assurance

The HR platform is covered by two automated backend suites — nineteen integration test files exercising the services against a live PostgreSQL instance, and a separate unit suite covering pure business logic such as statutory payroll calculation without a database dependency. Integration tests run sequentially because they share database state. The frontend is tested with component tests and headless browser end-to-end automation, with automated accessibility checking against WCAG rules. Both codebases enforce a zero-warning lint policy and strict type checking as separate gates.

Security testing is documented separately, comprising a penetration test report and a threat model, with rate limiting on authentication endpoints, security headers, environment-scoped CORS and schema validation at every endpoint.

The limitation to record is that this establishes **functional correctness, not usability**. That distinction is what §5.10 addresses, and the two should not be conflated: a system can be entirely correct and still be unusable.

5.3 Headline Result: Transfer versus Local

The two models defined in §3.2 were evaluated against the same Sri Lankan sample, with the results in Table 5.1.

Table 5.1 — Two-model comparison on the held-out Sri Lankan sample (n = 230, 33 positives)

Model	Features	ROC-AUC	PR-AUC
Transfer (trained internationally)	4	0.641	0.285
Local (cross-validated within Sri Lanka)	8	0.937	0.790

The headline contrast is large and it is the empirical contribution of this work. A model carrying knowledge from Saudi and Russian labour markets performs close to chance on Sri Lankan data, while a model fitted to Sri Lankan psychometric constructs performs strongly. This quantifies the research gap: **Sri Lanka needs its own HR data, and the absence of it cannot be substituted by international data on shared demographic features.**

Three qualifications must accompany that statement, and none of them is minor.

The two figures are not like-for-like. The transfer figure is genuine out-of-sample performance across national contexts. The local figure is cross-validated *within* the same 230 records. They answer different questions — "does knowledge transfer?" and "is there learnable signal here?" — and the gap between them is therefore not a clean experimental contrast between two competing models.

The transfer figure is unstable. The 0.641 headline is produced by the resampling configuration used in the training script at a single seed. Across five seeds that same configuration averages 0.718 with a standard deviation of 0.100 — ten AUC points of seed-to-seed variation. The alternative class-weighting configuration averages 0.828 with a standard deviation of 0.014. The transfer model's measured performance therefore depends more on the resampling choice and the random seed than on anything the data contains. A task with substantial signal does not swing nineteen AUC points on a preprocessing decision, and §5.5 shows what is actually driving the variation.

The outcomes are differently defined. As established in §3.2, the training data records observed attrition and the validation data records turnover intention. Part of the transfer model's weakness is attributable to this label shift rather than to national context, and the design cannot separate the two.

The local model's construct-level explanations are summarised in Figure 5.1.

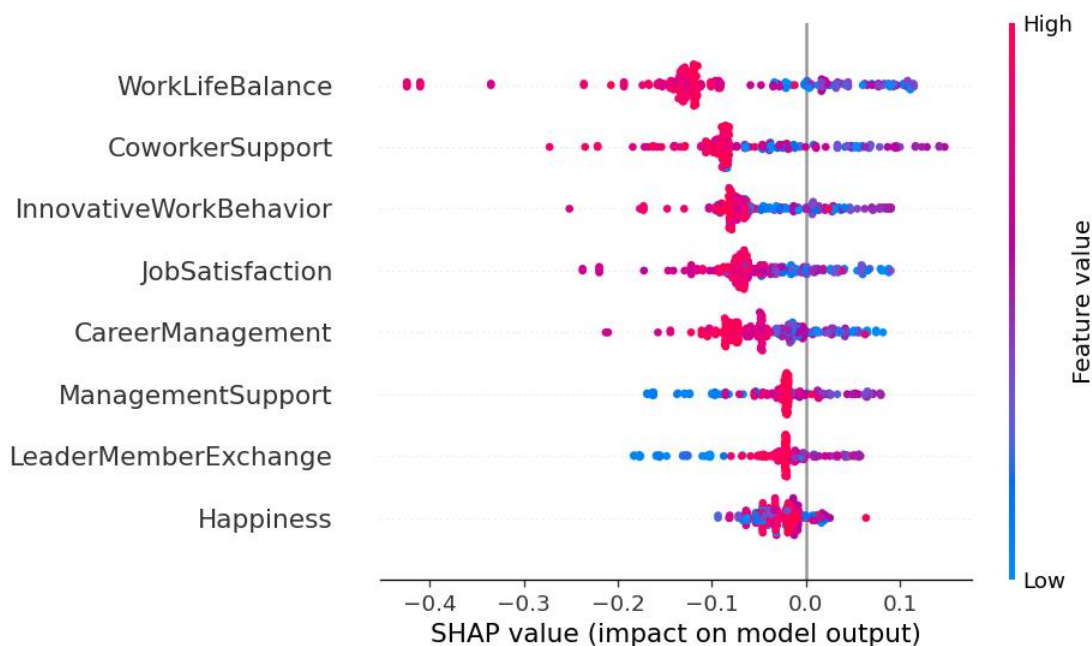


Figure 5.1 — SHAP summary for the local model, showing per-construct contributions to predicted turnover intention.

A note on the transfer model's feature importances is required. Age accounts for 0.612 of that model's importance, which the interim report described as age dominating the model. That

description should not be carried forward. As §5.8 shows, age in the Sri Lankan validation data takes only four distinct coded values with 88.7% of records at a single one; it is almost certainly a set of age-bracket midpoints rather than measured ages. A feature with four levels and one dominant mode cannot support a substantive claim about age effects, and the apparent dominance is an artefact of that coding.

5.4 Leakage and Common-Method Bias Audit

A ROC-AUC of 0.94 on 230 records with 33 positives invites the suspicion that something has leaked. The audit tested four possibilities; three are cleared and one is disclosed.

Imputation leakage: none, and structurally impossible. The construct data has a missing rate of exactly zero, so imputation is a no-op. Refitting with imputation confined inside each cross-validation fold reproduces the original result to four decimal places — 0.9366 either way, a difference of 0.000.

Item overlap between predictors and target: none. The turnover-intention composite is computed from four items that form a disjoint group from every predictor construct. Construct-to-target correlations range from -0.429 to -0.613 ; these reflect genuine association together with shared-method variance, but not the arithmetic self-correlation that item overlap would produce.

Calibration exposure: present, disclosed, and bounded. The audit found one path this section cannot clear. The synthetic-data calibration was fitted across all three sources, the Sri Lankan validation sample among them (§3.5), so validation records indirectly shaped 500 of the 2,820 training records. No Sri Lankan record entered model fitting, and the affected component contributes $+0.032$ ROC-AUC (§5.5), inside seed noise — but the exposure is real and is reported rather than argued away. Its direction favours the transfer arm, so it cannot account for that arm's failure.

Operating point: optimistic, and now corrected. The originally reported precision of 0.73 and recall of 0.82 were obtained at a threshold selected on the same predictions being scored. Selecting the threshold nested within the training folds gives the honest figure: **precision 0.58, recall 0.879**. Recall in fact improves; precision falls by fifteen points. The nested figure is the one reported throughout this dissertation.

The 0.94 therefore survives, with an honest confidence interval. Bootstrap resampling gives a 95% interval of [0.883, 0.982], per-fold standard deviation is 0.047, and the Brier score is 0.056, indicating well-calibrated probabilities rather than merely good ranking.

What the audit **cannot** rule out is common-method bias in the strict sense [38]. Predictors and outcome are collected from the same respondent, in the same instrument, at the same moment. Some portion of the 0.94 reflects consistency in how individuals answer self-report scales rather than a predictive relationship between constructs and behaviour. This is a limitation of the source instrument, not of the modelling, and it cannot be resolved without behavioural outcome data.

5.5 Synthetic Data Ablation

RQ3 asks whether a hybrid dataset — real international attrition data combined with calibrated synthetic Sri Lankan data — can produce a usable classifier for a low-volume data environment. Answering it requires testing two assumptions the data strategy rests on: that the synthetic component contributes anything, and that the 2.0 and 0.5 sample weights are doing work. Six conditions were run across five seeds, and Table 5.2 reports them.

Table 5.2 — Synthetic-data ablation on the transfer model (mean ROC-AUC across five seeds)

Condition	ROC-AUC	SD
Class weighting, real 2.0 + synthetic 0.5 (current recipe)	0.821	0.020
Class weighting, equal weights	0.828	0.014
Class weighting, real data only	0.788	0.031
Class weighting, synthetic only	0.526	0.103
SMOTETOMEK with synthetic (headline path)	0.718	0.100
SMOTETOMEK, real only	0.655	0.090

Three conclusions follow. Synthetic augmentation produces a small positive effect of +0.032 ROC-AUC under class weighting, at the edge of seed noise. The 2.0 and 0.5 weighting is

immaterial — a difference of -0.007 against equal weights, well inside noise, so the weighting scheme is not doing the work it was assumed to do. And synthetic data alone performs at 0.526, effectively chance, confirming that it functions as augmentation and could not substitute for real data.

The more consequential finding came from decomposing the transfer signal by feature group. Age and gender together score **0.457** — below chance. Job satisfaction and work-life balance together score **0.825**, essentially the full four-feature performance of 0.828.

Whatever the transfer model achieves comes entirely from satisfaction items, not from demographics. And those satisfaction items are, on the Sri Lankan side, self-report collected in the same instrument as the intention outcome. The transfer model is therefore not demonstrating cross-cultural transfer of attrition knowledge; it is partly re-detecting same-source correlation between satisfaction and intention. This reframes the headline: the transfer result is weaker than 0.64 makes it appear, not stronger. Figure 5.2 shows the full ablation across all six conditions and five seeds.

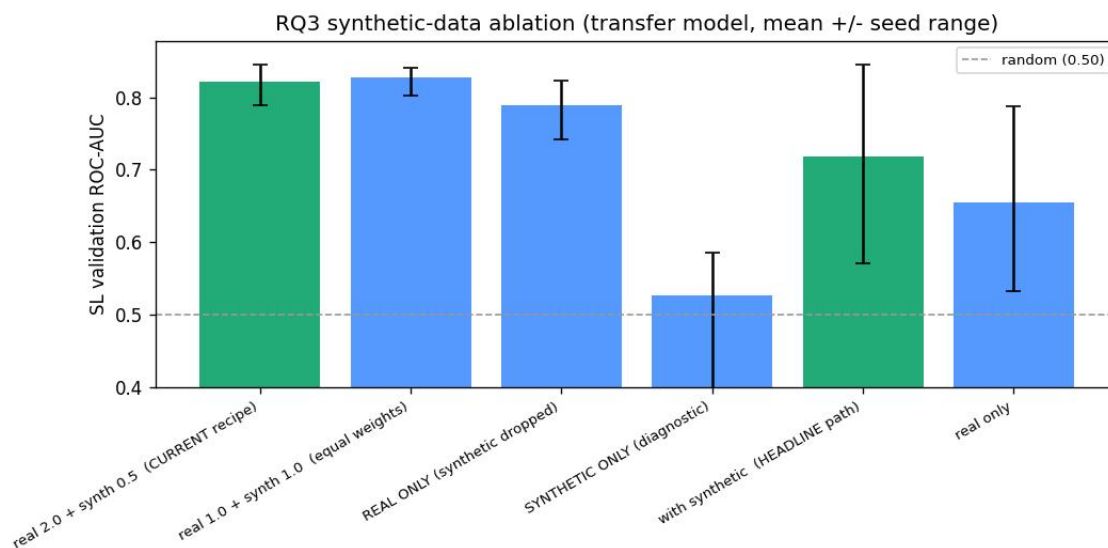


Figure 5.2 — Synthetic-data ablation across six conditions and five seeds.

5.6 Estimator Justification

Random Forest was compared against logistic regression and gradient boosting on identical data, folds and seeds. Table 5.3 reports that comparison.

Table 5.3 — Estimator comparison (mean ROC-AUC across five seeds)

Arm	Random Forest	Gradient Boosting	Logistic Regression
Local (8 constructs)	0.937	0.853	0.823
Transfer, class weighting	0.821	—	0.854

On the local arm Random Forest wins outright: +0.084 over the next best, a difference of 7.79 standard deviations against seed noise, and it is also the best calibrated of the three (Brier 0.061 against 0.166 for logistic regression). The estimator choice is vindicated exactly where the strong result lives.

On the transfer arm it does not win. Logistic regression is better by 0.033 and roughly ten times more stable. This is reported rather than omitted, and it corroborates §5.5: a four-feature task on which a linear model matches an ensemble is a task without much structure to find. The retention of Random Forest for the deployed local model is justified on performance; its use in the transfer arm is a consistency choice, not a performance one, and is stated as such. Figure 5.3 plots both arms against both resampling paths.

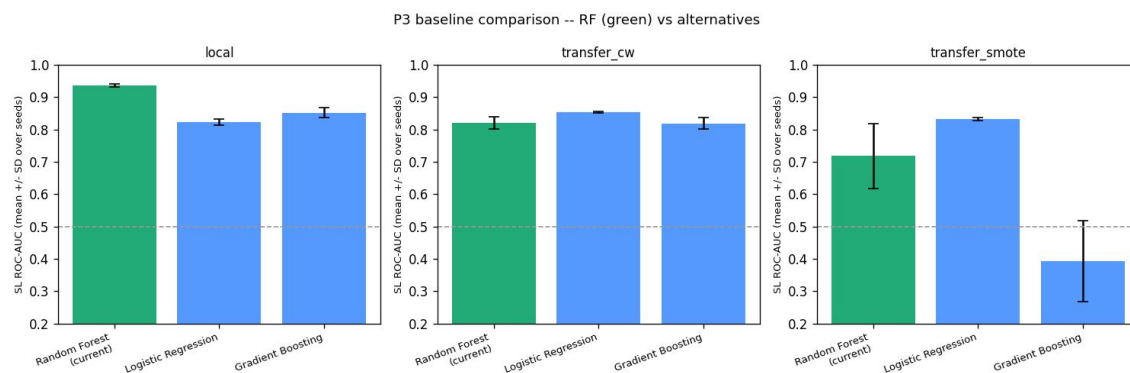


Figure 5.3 — Estimator comparison across both arms and both resampling paths.

5.7 Threshold Sensitivity

The binary label depends on a cut at 3.5 on the intention composite. If the headline contrast were an artefact of that choice, the finding would be fragile. Table 5.4 reports the contrast at alternative cut-points.

Table 5.4 — Contrast at alternative binarisation cut-points

Cut	Positives	Local	Transfer (CW)	Contrast
≥ 3.0	51	0.908	0.777	+0.131
≥ 3.5 (adopted)	33	0.937	0.821	+0.116
≥ 4.0	12	—	—	indicative only

The contrast survives every cut tested, ranging from +0.095 to +0.218 depending on cut and resampling path. Only the 3.0 and 3.5 cuts clear the twenty-positive reliability bar adopted for this analysis; at 4.0 only twelve positive cases remain and the bootstrap interval widens accordingly, so that cut is reported as indicative and excluded from the conclusion.

One disclosure is owed: **3.5 is also where the local model performs best**, which means the adopted threshold flatters the strongest number. The contrast does not depend on it, but the specific figure of 0.937 does. Figure 5.4 shows performance and contrast across every cut-point tested.

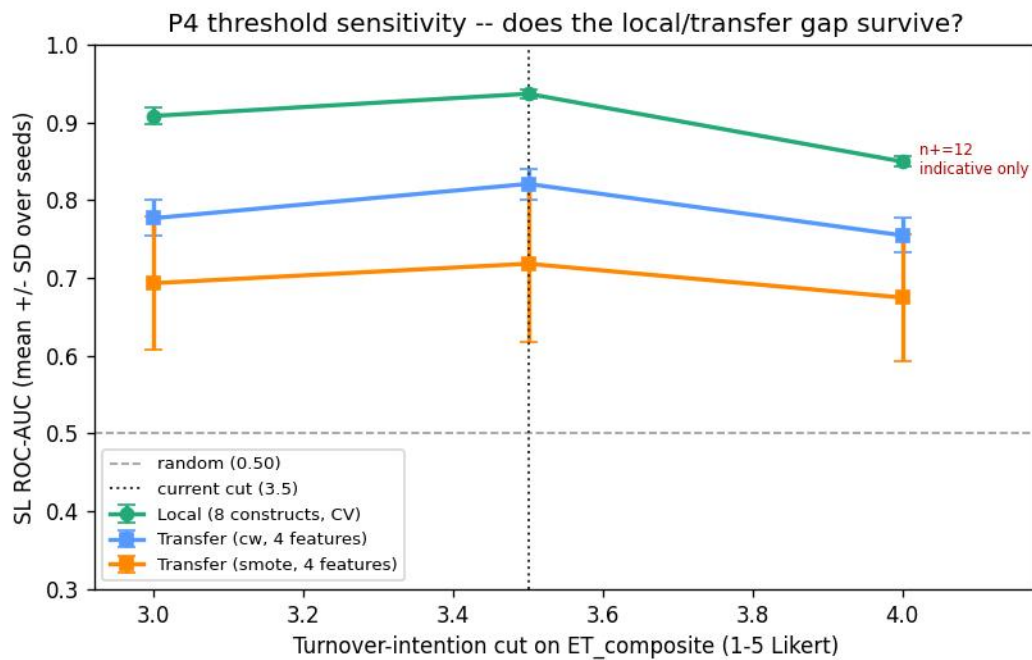


Figure 5.4 — Model performance and contrast across binarisation cut-points.

5.8 Fairness Audit

Fairness cannot be fully validated on this dataset, and the reason is structural rather than an oversight. Reporting why is more useful than reporting a table that the data cannot support.

The sample contains 73 women, of whom 2 are positive cases, and 157 men, of whom 31 are. Age takes four distinct coded values — 25, 35, 45 and 52 — with 88.7% of records at 25. A subgroup ROC-AUC computed on two positive cases is meaningless, and an age-band analysis on a variable with one dominant level is not supportable. The audit therefore refuses to compute those figures and marks the subgroups unevaluable rather than printing numbers with a caveat elsewhere. Table 5.5 reports subgroup performance only where the data supports it.

Table 5.5 — Subgroup performance where the data supports it

Group	n	Positives	ROC-AUC	Recall
All	230	33	0.943	0.818
Gender = male	157	31	0.941	0.807
Gender = female	73	2	unevaluable	—
Age = 25	204	27	0.943	0.778
Age > 25	26	6	unevaluable	—

Where performance is evaluable it is consistent with the overall figure. Selection-rate parity was assessed against the four-fifths rule, and **both attributes fail it — but the two failures mean opposite things**, which is why the raw ratios alone would mislead.

For gender, the selection-rate ratio is 0.416, below the 0.8 threshold. The **base-rate** ratio, however, is 0.139: the underlying outcome disparity is far larger than the disparity the model produces. Women are flagged at three times their own base rate, men at one time theirs. Under differing base rates, four-fifths parity and calibration are mathematically incompatible [41], [42]; this is a documented trade-off to argue, not a defect to remove. For age, the selection ratio of 0.462 sits *below* the base-rate ratio of 0.574, meaning the model amplifies rather than narrows the existing disparity — a genuine flag, though on a subgroup of 26 with 6 positives and therefore indicative only.

Two further tests inform the deployment position. A proxy test asked whether gender can be recovered from the eight constructs the deployed model actually uses: cross-validated ROC-AUC **0.655** — weak but non-zero. The claim "the local model excludes protected attributes" is therefore *mostly* true and must be stated that way. And a drop-and-test on the transfer model found that removing age and gender changes performance by **+0.004** — that is, removing the protected attributes costs nothing and marginally helps, consistent with the finding in §5.5 that they score below chance alone.

The recommendation is therefore unambiguous: age and gender should be dropped from the transfer model. There is no accuracy argument for retaining them. Subgroup validation is treated as an explicit precondition of real deployment, not as a solved problem. Figure 5.5 shows the subgroup structure alongside the adverse-impact analysis.

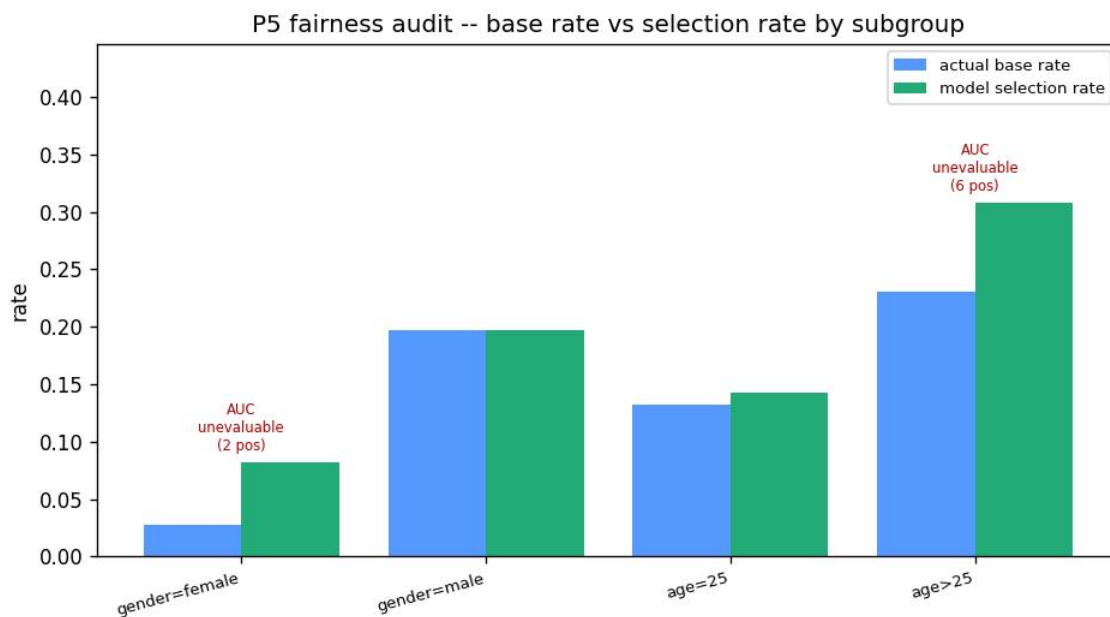


Figure 5.5 — Subgroup structure and adverse-impact analysis.

5.9 Objective O3 — Recall Above 80%

The target was recall above 80% on the attrition class, chosen because the costly error in early warning is the employee who leaves unflagged.

The objective is met by the local model at its honest operating point: recall 0.879, precision 0.580. This is the nested threshold from §5.4, not the optimistic one.

The transfer model must not be credited with meeting it. On the Sri Lankan sample it achieves recall of 1.000 at the default threshold — by flagging 86% of the workforce, at a precision of 0.167. Perfect recall obtained by flagging almost everybody is not early warning; it is noise. This is precisely why §3.8 specifies that recall is never reported alone, and it is the clearest illustration in this work of why a single metric cannot carry an evaluation.

5.10 Objective O5 — Usability

The planned System Usability Scale study was not conducted. No participants were recruited and the remaining project timeline did not permit recruitment. The objective is therefore **unmeasured — neither met nor refuted** — and is reported as such rather than quietly dropped.

Part A — the protocol as designed. The instrument is the standard ten-item System Usability Scale [33], administered after task completion to five to ten participants matched to the SME HR-manager persona. Four task scenarios were specified: locate the dashboard after login; identify the three highest-risk employees; explain why a given employee is flagged; and generate a month's payroll. Responses are scored to the 0–100 composite and interpreted against established adjective bands, on which a score above 80 corresponds to "excellent" and 68 represents the industry average [34]. The protocol is fully specified and directly repeatable by anyone continuing this work.

Part B — heuristic evaluation. In place of user-based measurement, the interface was assessed by expert inspection against Nielsen and Molich's ten usability heuristics [35], applied across the same four task scenarios, with each observation assigned a severity on Nielsen's 0–4 scale [47].

Procedure and evaluators. The inspection was conducted on 23–24 August 2026 against the development deployment, model bundle 2026.07.31. Two evaluators contributed, and the division of labour is stated because it bears on how the findings should be weighted. **E1** is the author, a human evaluator with complete knowledge of the system. **E2** is an AI assistant driving a scripted browser session, which walked the four scenarios, captured a screenshot for every observation, and instrumented latency and interaction counts. E2 proposed a candidate finding set; E1 verified each against the captured evidence and contributed the findings E2 could not produce. Supporting measurements — response latencies, click paths, console output — are instrumented; the assignment of heuristics and severities is E1's.

The distinction matters because an automated evaluator reasons about principles against captured evidence but does not *experience* an interface. It cannot report confusion, hesitation or the feeling of not knowing where to begin, and heuristic evaluation was designed around evaluators who can. The single most consequential finding below (F13) came from E1 and could not have come from E2.

Table 5.6 — Heuristic evaluation findings

#	Heuristic	Screen	Observation	Sev.
F1	H10 Help & documentation	Attrition risk card	All eight constructs at the neutral midpoint return 77.5% probability, band HIGH against a disclosed 30.2% threshold. The card explains what the number is, but not what a neutral input should be expected to produce, nor what a manager should do with a HIGH band.	4
F2	H9 Error recovery	Login	A network-layer failure renders "Login failed. Check your credentials and try again." The credentials were correct; the message misattributes an infrastructure fault to user error.	3
F3	H7 Flexibility &	Employees, dashboard,	No risk column, sort, filter, dashboard tile,	3

	efficiency	navigation	navigation entry or manager-side aggregate exists. Ranking n employees requires n page loads and n manual predictions. Scenario 2 cannot be completed as specified.	
F4	H5 Error prevention	Risk card, Pulse Check	Both instruments default every item to the midpoint, so an untouched form submits as a complete neutral response. Neither distinguishes "answered neutral" from "not answered".	3
F5	H6 Recognition vs recall	Employees	Absent a comparative view, the manager must hold multiple probabilities in working memory to rank them.	3
F13	H10 Help & documentation	Whole system	A first-time user is not told where to begin. There is no onboarding, guided first run, or entry-point cue; the	3

			interface presents its full navigation and assumes orientation. Reported by E1 on first-use simulation.	
F6	H4 Consistency & standards	Attrition risk card	The same eight constructs are named two ways on one card: sliders read "Work-life balance", "Coworker support"; the results panel beneath reads WorkLifeBalance, CoworkerSupport.	2
F7	H2 Match to real world	Attrition risk card	Card copy calls the Pulse Check "planned" and cites "future source: Dialogflow Pulse Check". The Pulse Check is deployed and Dialogflow was deliberately not used (§4.7).	2
F8	H1 Visibility of status	Attrition risk card	First prediction takes ≈ 20 s against 523–1,092 ms warm. The control shows "Scoring..." and disables, so feedback exists, but no progress indication or advance warning	2

			is given.	
F9	H6 Recognition vs recall	Payroll	All row checkboxes carry the accessible name "Select row" with no employee identity.	2
F10	H8 Aesthetic & minimalist	Pulse Check	The scale anchor "Disagree" is clipped by its container on all 16 items.	1
F11	H3 User control & freedom	Payroll	Generation runs immediately on click with no confirmation, mitigated by the idempotency noted below.	1
F12	H2 Match to real world	Dashboard	A page titled "company-wide overview" devotes a central panel to the viewer's personal leave balances.	1

Severity distribution. Of 13 findings: one catastrophe (4), five major (3), four minor (2), three cosmetic (1). The distribution is unusually top-heavy for an interface of this maturity, and the reason is visible in the table — the severe findings cluster on the predictive feature rather than on the HR platform, which is the newer and less exercised part of the system.

The three leading issues.

First, the risk output can be acted on without being understood (F1). Neutral inputs produce a HIGH band because the operating point is deliberately tuned for recall (§5.7), which is defensible as modelling and dangerous as interface. A manager who enters midpoint values for

an employee they know little about receives an alarming result and no interpretive frame. The threshold is disclosed, which is more than most systems offer, but a disclosed threshold is not an explanation of what the number means.

Second, the system's primary use case is unsupported (F3). The framework exists to give a manager early warning about their workforce, yet the interface answers only "what is this employee's risk?" and never "who should I worry about?". This is a scope gap rather than a defect of execution — the risk card was built as a per-employee explainer — but the effect on the stated purpose is the same, and it is carried into Chapter 6 as future work.

Third, a first-time user is not told where to begin (F13). This finding connects directly to the adoption evidence in §2.5. Ilyas et al. [31] find SME adoption of HR analytics driven by effort expectancy, facilitating conditions and trust rather than analytical capability, and Bahanni et al. [32] describe how weak internal capability compounds external constraint in developing-economy firms. An interface that assumes orientation places its cost precisely on effort expectancy — the variable the literature identifies as decisive for exactly this population. Taken together, F1, F3 and F13 indicate that the barrier to this system being adopted is not its accuracy.

Behaviour that met the heuristics. Reporting only defects would misrepresent the interface. Payroll generation is idempotent, reporting "Created: 0 | Skipped (already exists): 1" on a repeat run rather than duplicating financial records. The generation control states its own scope and is disabled at zero selection. The decision threshold is disclosed alongside the probability. The turnover-*intention* caveat appears in the risk card at the point of use rather than in documentation. The Pulse Check states its confidentiality contract to the respondent before any question is asked. Three properties tested specifically were confirmed: keyboard-only completion of the login-to-risk path is possible; the layout remains usable at narrow viewport widths; and **an employee cannot view their own risk score**, which verifies the privacy-by-design claim made in §4.7.

Three limitations apply to this substitution regardless of outcome, and none is repaired by the findings above.

Heuristic evaluation is an **inspection method, not a measurement**. It surfaces usability defects but yields no satisfaction score, so it cannot answer the "above 80" target in its own terms. **Objective O5 therefore remains partly undischarged**, and Table 5.6 should not be read as a substitute result.

Evaluator independence is weak. Nielsen and Molich recommend three to five evaluators [35], and this inspection had two, of whom the only human is the system's own author. Developer-evaluators systematically under-report problems they have internalised, and no procedure here corrects for that. The second evaluator does not share the author's mental model, which is some mitigation, but it is not independence.

The second evaluator is not a usability expert. E2 is an AI assistant, and its contribution should be understood as systematic coverage rather than expertise: it walked every scenario, checked every screen against all ten heuristics without fatigue, and produced a screenshot for each claim. What it cannot do is experience the interface, and one of the two highest-value findings (F13) is precisely of that kind. An inspection conducted by E2 alone would have missed it, and the same is likely true of other problems that only manifest as felt difficulty. The findings below the top two severities should therefore be read as reliable in what they assert and incomplete in what they omit.

5.11 Objective O2 — Operational Cost Below LKR 10,000 per Month

The objective is met, with approximately 2.5 times headroom: LKR 4,050 per month on the most conservative scenario evaluated.

As established in §3.10, project-level billing could not answer this question because the deployment shares a cloud project with four unrelated systems. Cost was therefore measured per resource over a four-month window and priced against published rates, under the four scenarios in Table 5.7.

Table 5.7 — Attributed monthly cost against the LKR 10,000 target

Scenario	USD	LKR	Result
Attributed share, free tier applied	7.47	2,256	Pass
Attributed share, free tier ignored	7.79	2,353	Pass
Single-tenant SME, free tier applied	13.09	3,953	Pass

Single-tenant SME, free tier ignored	13.41	4,050	Pass
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The headline is the last row, because it makes no use of a free-tier allowance that is granted per billing account rather than per service, and because it charges the whole database instance to this system as a dedicated deployment would.

Three findings are more interesting than the pass itself. **Compute is not the cost:** all three services total approximately LKR 82 per month and fall entirely within the free tier, and the inference service consumed **150 billable instance-seconds across four months** — scale-to-zero behaving exactly as §4.5 claims. **The always-on database dominates** at LKR 2,828 per month, roughly 70% of the total: the one component that cannot scale to zero is the one that costs money, which is the architectural lesson of the cost study. And **container image storage is a hidden line item** at LKR 1,140 per month for 37.7 GB of accumulated build artefacts — real, but largely removable with a retention policy.

Three limitations qualify the result. Unit prices are published rate-card figures rather than invoiced amounts. The apportionment in the attributed scenarios rests on an assumption about shared-database usage, which the single-tenant scenario avoids. Most importantly, **the measured usage reflects development traffic, not a live fifty-employee workload**; the measured and modelled figures are reported separately and neither should be read as the other. Figure 5.6 shows attributed cost by scenario together with the breakdown by service.

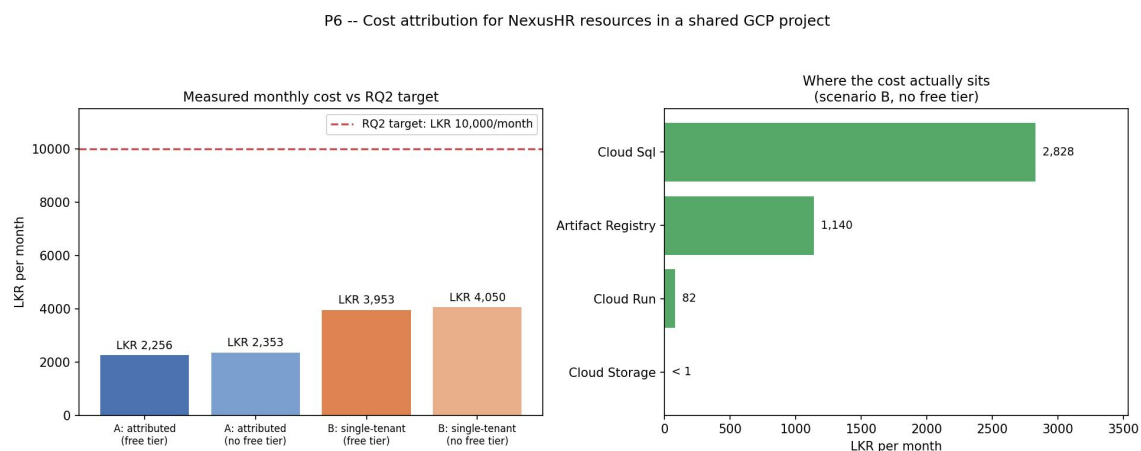


Figure 5.6 — Attributed monthly cost by scenario, and the cost breakdown by service.

5.12 Threats to Validity

Construct validity. The Sri Lankan outcome is turnover *intention*, not observed departure. Intention is an established antecedent of turnover [16] but the two are not equivalent, and every claim made about the local model is a claim about intention. Relatedly, predictors and outcome share a single self-report instrument, so common-method variance [38] inflates the local result by an amount this design cannot quantify.

Internal validity. Imputation leakage and item overlap were tested and excluded (§5.4), and the operating point was corrected. The residual threat is that the local figure is a within-sample cross-validated estimate on 230 records with 33 positives; the bootstrap interval of [0.883, 0.982] is wide, and small-sample optimism cannot be fully excluded.

External validity. The validation sample is drawn from Sri Lankan *startup* professionals, which is not the SME population the framework targets. The training sources are Saudi and Russian, chosen for availability rather than similarity. Cost is measured on development traffic in a shared project. None of these generalises without qualification.

Conclusion validity. All comparisons are averaged across five seeds and reported with dispersion, and differences are stated against seed noise rather than as bare point estimates. Where the data cannot support an inference — subgroup fairness, the 4.0 cut-point — the analysis declines to compute it rather than reporting a fragile number.

Fairness and deployment. Subgroup performance is unvalidated for women and for employees over 25 (§5.8), and gender is partly recoverable from the deployed model's inputs at ROC-AUC 0.655. The system should not be deployed on real employees without subgroup validation on a sample that can support it.

Artifact evidence. The Pulse Check was implemented and functionally verified but never populated with real employee responses (§4.7); its short-form reliability is consequently unassessed, and no claim about real employee sentiment is made anywhere in this work.

Chapter 6 — Conclusions and Future Work

6.1 Answers to the Research Questions

Each question is answered first in a sentence, then elaborated. The wording is that of the research questions as stated in the interim report; §1.7 records how RQ1 and RQ3 evolved from the approved proposal and why.

RQ1 — what are the most significant predictors of attrition for the Sri Lankan SME context, and to what extent do attrition patterns transfer from international data?

The significant predictors are psychometric rather than behavioural, and attrition patterns transfer barely at all.

Within the Sri Lankan sample, all eight psychometric constructs relate negatively to turnover intention, with job satisfaction the strongest ($r = -0.613$), followed by career management (-0.571) and innovative work behaviour (-0.559). Together they support a strongly discriminating model (ROC-AUC 0.937).

Transfer, however, is close to absent. The cross-context model reaches 0.641 at the reported configuration and is unstable — 0.718 ± 0.100 on one resampling path against 0.828 ± 0.014 on another, meaning its measured performance depends more on a preprocessing choice than on the data. Decomposition then removes most of what remains: age and gender together score **0.457, below chance**, while satisfaction items alone score 0.825 — effectively the entire model. Those satisfaction items are, on the Sri Lankan side, self-reported in the same instrument as the outcome. What looks like transferred knowledge is substantially same-source correlation. **The honest answer is that international attrition patterns do not meaningfully transfer to the Sri Lankan context on the features the two bodies of data share.**

RQ2 — to what extent can a serverless architecture reduce operational cost relative to persistent infrastructure, and can it stay within LKR 10,000 per month?

The budget question is answered decisively; the comparative question only partially.

Measured operational cost is **LKR 4,050 per month** on the most conservative scenario, roughly 2.5 times inside the ceiling. The mechanism is demonstrated rather than asserted: the inference service consumed 150 billable instance-seconds across four months, and all compute together

accounts for about LKR 82 per month, while the one component that cannot scale to zero — the always-on database — accounts for roughly 70% of total spend. That distribution is itself the answer to how serverless reduces cost: it removes the idle cost of the components that can be made stateless, and leaves untouched those that cannot.

The comparison against *persistent* infrastructure is, however, argued architecturally rather than measured. The planned controlled comparison across alternative deployment topologies was not carried out, so the reduction relative to an always-on deployment is inferred from the cost structure rather than benchmarked. This is a genuine incompleteness in the answer to RQ2.

RQ3 — can a hybrid dataset combining real international data with calibrated synthetic data produce a usable classifier for a low-volume Sri Lankan environment?

No.

The hybrid approach produced a classifier, but not a usable one. On Sri Lankan data the resulting model achieves an F1 of approximately 0.29, and reaches high recall only by flagging 86% of the workforce at a precision of 0.167. Against the criterion originally set in the proposal — $F1 \geq 80\%$ — it fails, and so does every configuration tested. The ablation explains why: synthetic data alone performs at 0.526, indistinguishable from chance; synthetic augmentation contributes +0.032 ROC-AUC, at the edge of seed noise; and the 2.0 and 0.5 sample weights make a difference of -0.007 , meaning the weighting scheme central to the data strategy is doing no measurable work.

The usable classifier in this project came from local data alone, not from the hybrid. **This is the central negative result of the work, and it is more valuable than a positive one would have been:** it establishes empirically that synthetic augmentation and international borrowing cannot substitute for local data collection in this domain, which is precisely the gap the project set out to examine.

6.2 Contributions

Four contributions are claimed.

First, and principally, **an empirical demonstration that HR attrition models do not transfer across national contexts on shared demographic features**, while a locally-fitted psychometric

model performs strongly (0.937 against 0.641). The gap was shown to survive leakage correction, estimator substitution and alternative binarisation of the outcome. More informatively, decomposition showed that age and gender score *below chance* (0.457) and that the transfer model's apparent skill comes almost entirely from satisfaction items which are, on the validation side, same-source self-report. The transfer result is therefore weaker than its headline suggests — which strengthens rather than weakens the argument for local data collection.

Second, **a cost-viable serverless reference architecture** for SME-scale predictive HR analytics, measured rather than estimated, with the finding that compute is not the cost: the always-on database accounts for roughly 70% of operational spend while inference is effectively free.

Third, **a working data-capture mechanism** — the Pulse Check — that produces the model's own input constructs inside the product, closing the gap between a model that performs well in a study and one that could be operated.

Fourth, **the reporting of negative and infeasible results** rather than their suppression: the transfer failure, the immateriality of the sample-weighting scheme, and the structural impossibility of subgroup fairness validation on this sample.

6.3 Limitations

The limitations are set out fully in §5.12 and are summarised here without softening.

The Sri Lankan outcome is turnover **intention**, not observed departure; every claim about the local model is a claim about intention. Predictors and outcome come from a single self-report instrument, so common-method variance inflates the local result by an amount this design cannot quantify. The local figure is a within-sample cross-validated estimate on 230 records with 33 positives, with a wide bootstrap interval of [0.883, 0.982]. The validation sample is drawn from startup professionals rather than the SME population targeted, and the training sources were selected for availability rather than similarity. The validation sample also contributed to the synthetic-data calibration (§3.5), so it is not untouched by the training pipeline, though the exposure is aggregate-level and works against the transfer arm rather than for it.

Fairness is unvalidated for women and for employees over 25 because the sample cannot support it — two positive cases and six respectively — and gender remains partly recoverable from the

deployed model's inputs at 0.655. The system should not be used on real employees without subgroup validation on an adequate sample.

Usability was not *measured*: the System Usability Scale study did not run, so the "above 80" target is neither met nor refuted and objective O5 remains only partly discharged. A heuristic inspection was substituted and did return findings (§5.10), but it was conducted by two evaluators of whom the only human is the system's author, which is materially weaker than the independence the method assumes. Cost was measured on development traffic in a shared project, not on a live fifty-employee workload, and the comparison against persistent infrastructure was argued rather than benchmarked. The Pulse Check, while functional, was never populated with real responses, so its short-form reliability is unassessed.

Objective O1 required compliance with the Personal Data Protection Act. What this work establishes is a compliant **design** rather than a compliance **result**: identifying data is stripped at the database layer, access is role-gated and operations are audit-logged, but the automated pre-transfer masking specified in the architecture was never provisioned, and — because no real employee data was ever processed — none of these controls has been exercised on live personal data.

6.4 Future Work

Obtain observed attrition data from a partner SME. This is the single highest-value next step. Behavioural outcome data would dissolve the label-shift confound that limits the transfer comparison and would allow the local model to be evaluated against departures rather than intentions. It is also the least controllable, requiring an organisational partner willing to share outcome records.

Refit the synthetic calibration on the international sources alone. The calibration step currently includes the Sri Lankan validation sample (§3.5), which puts an indirect path between the held-out data and the training set. The effect is bounded and favours the arm this work reports as failing, so no conclusion here depends on it — but the pipeline should be corrected before any of these results are built on, and the transfer figures regenerated from a calibration fitted only on the Saudi and Russian sources.

Validate the Pulse Check short form. Once roughly fifty real responses exist, the internal consistency of the two-item construct pairs can be assessed, which would either defend the short-form design empirically or justify lengthening it.

Conduct the usability study. The protocol is fully specified in §5.10 and is directly executable with independent evaluators; completing it would close the one objective this work leaves unanswered.

Build the workforce risk view, and an onboarding path. The heuristic inspection found that the system's primary question — *which of my employees should I worry about?* — has no interface (§5.10, F3), and that a first-time user is given no place to start (F13). Both are additions rather than corrections, and both matter more than they appear: §2.5 identifies effort expectancy and facilitating conditions, not accuracy, as what determines whether an SME adopts a tool of this kind. A ranked risk view is also close to free once Pulse Check responses exist, since the predictions are already cached per employee per week (§4.7).

Validate fairness on an adequate sample. Subgroup performance should be treated as a deployment precondition. On the evidence available, age and gender should also be dropped from the transfer model outright, since removing them costs nothing.

Reduce the dominant cost. Since the always-on database accounts for most of the operational spend, a serverless or scale-to-zero database tier would be the obvious next architectural target — as would a container image retention policy, which alone accounts for LKR 1,140 per month.

6.5 Reflection

The most useful thing this project produced was not the strong model but the weak one. The original expectation was that international training data could substitute for absent local data; the evidence says it cannot, and says so in a way that is now robust to several attempts to explain it away. Reporting that clearly — including the discovery that the transfer model's residual skill was largely same-source correlation rather than transferred knowledge — proved more valuable than the headline accuracy figure it sits beside.

The second lesson was methodological. Several claims made confidently at the interim stage did not survive scrutiny: an operating point tuned on the data it was scored against, a sample-weighting scheme that turned out to be immaterial, and a statement about age dominating the

transfer model that was an artefact of how age had been coded. Each was found by writing a script specifically to attack a number rather than to produce one. That practice — one script per question, a machine-readable report per script, and a refusal to print a figure the data could not support — is what makes the limitations in this dissertation credible rather than decorative.

References

- [1] Department of Census and Statistics, "Sri Lanka Labour Force Survey — Annual Report 2023," Ministry of Finance, Sri Lanka, 2023.
- [2] S. Kirupananthan, M. Kayathiry, and S. Niroshan, "Exploring Challenges for Artificial Intelligence Readiness in SMEs in Sri Lanka," *Vavuniya Journal of Business Management*, vol. 4, no. 1, 2024.
- [3] B. Charney, "HR Software Pricing Guide: How Much Does HR Software Cost?," People Managing People, 2024.
- [4] E. Jonas et al., "Cloud programming simplified: A Berkeley view on serverless computing," arXiv:1902.03383, 2019.
- [5] Information and Communication Technology Agency of Sri Lanka (ICTA), "Sri Lanka Digital Economy Report 2023," ICTA, Colombo, 2023.
- [6] Ministry of Technology, "Artificial Intelligence in Sri Lanka: White Paper," Government of Sri Lanka, 2024.
- [7] D. G. Allen, *Retaining Talent: A Guide to Analyzing and Managing Employee Turnover*. Alexandria, VA: SHRM, 2008.
- [8] Sri Lanka Bureau of Foreign Employment, "Annual Statistical Report of Foreign Employment 2023," SLBFE, Colombo, 2023.
- [9] I. H. Sarker, "Machine Learning: Algorithms, Real-World Applications and Research Directions," *SN Computer Science*, vol. 2, no. 3, pp. 1–21, 2021.
- [10] R. Punnoose and P. Ajit, "Prediction of Employee Turnover in Organizations using Machine Learning Algorithms," *Int. J. Advanced Research in Artificial Intelligence*, vol. 5, no. 9, pp. 22–26, 2016.
- [11] M. Ribeiro, K. Grolinger, and M. A. M. Capretz, "MLaaS: Machine Learning as a Service," in *Proc. IEEE ICMLA*, 2015, pp. 896–902.

- [12] D. Angrave, A. Charlwood, I. Kirkpatrick, M. Lawrence, and M. Stuart, "HR and Analytics: Why HR is Set to Fail the Big Data Challenge," *Human Resource Management Journal*, vol. 26, no. 1, pp. 1–11, 2016.
- [13] Parliament of Sri Lanka, "Personal Data Protection Act, No. 9 of 2022," *Government Gazette of the Democratic Socialist Republic of Sri Lanka*, 2022.
- [14] L. Kanchana and R. Jayathilaka, "Factors impacting employee turnover intentions among professionals in Sri Lankan startups," *PLOS ONE*, vol. 18, no. 2, e0281729, 2023, doi: 10.1371/journal.pone.0281729.
- [15] W. H. Mobley, "Intermediate linkages in the relationship between job satisfaction and employee turnover," *Journal of Applied Psychology*, vol. 62, no. 2, pp. 237–240, 1977.
- [16] R. W. Griffeth, P. W. Hom, and S. Gaertner, "A Meta-Analysis of Antecedents and Correlates of Employee Turnover: Update, Moderator Tests, and Research Implications for the Next Millennium," *Journal of Management*, vol. 26, no. 3, pp. 463–488, 2000.
- [17] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [18] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [19] IBM, "IBM HR Analytics Employee Attrition & Performance Dataset," Kaggle, 2017.
- [20] W. Li, "A transformer-based deep learning framework to predict employee attrition," *PeerJ Computer Science*, vol. 9, e1570, 2023, doi: 10.7717/peerj-cs.1570.
- [21] M. S. Pavithran and S. M. Vadivel, "Explainable attrition risk scoring for managerial retention decisions in human resource analytics," *Frontiers in Big Data*, vol. 8, 1699561, 2025, doi: 10.3389/fdata.2025.1699561.
- [22] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.
- [23] S. M. Lundberg et al., "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, pp. 56–67, 2020, doi: 10.1038/s42256-019-0138-9.

- [24] K. Mohiuddin, M. A. Alam, M. M. Alam, P. Welke, M. Martin, J. Lehmann, and S. Vahdati, "Retention Is All You Need," arXiv:2304.03103, 2023.
- [25] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, pp. 206–215, 2019.
- [26] J. G. Moreno-Torres, T. Raeder, R. Alaiz-Rodríguez, N. V. Chawla, and F. Herrera, "A unifying view on dataset shift in classification," *Pattern Recognition*, vol. 45, no. 1, pp. 521–530, 2012, doi: 10.1016/j.patcog.2011.06.019.
- [27] R. Fischer and A. Mansell, "Commitment across cultures: A meta-analytical approach," *Journal of International Business Studies*, vol. 40, no. 8, pp. 1339–1358, 2009, doi: 10.1057/jibs.2009.14.
- [28] Google Cloud, "Vertex AI: Train and Deploy ML Models," Google LLC, 2024.
- [29] G. Adzic and R. Chatley, "Serverless computing: economic and architectural impact," in *Proc. 11th Joint Meeting on Foundations of Software Engineering (ESEC/FSE)*, Paderborn, Germany, 2017, pp. 884–889, doi: 10.1145/3106237.3117767.
- [30] A. Eivy and J. Weinman, "Be wary of the economics of 'serverless' cloud computing," *IEEE Cloud Computing*, vol. 4, no. 2, pp. 6–12, 2017.
- [31] M. Ilyas, W. Alam, and A. Ahmad, "Breaking barriers: driving HR analytics adoption in small and medium-sized enterprises," *Evidence-based HRM: a Global Forum for Empirical Scholarship*, vol. 14, no. 1, pp. 171–191, 2026, doi: 10.1108/EBHRM-01-2024-0015.
- [32] K. Bahanni, Y. Fujimoto, and E. E. T. Bolt, "Contextual challenges in HR analytics adoption: an open systems perspective from HR managers in a developing country," *Personnel Review*, vol. 55, no. 2, pp. 578–598, 2026, doi: 10.1108/PR-04-2024-0375.
- [33] J. Brooke, "SUS — A Quick and Dirty Usability Scale," in *Usability Evaluation in Industry*. London: Taylor & Francis, 1996, pp. 189–194.

- [34] A. Bangor, P. T. Kortum, and J. T. Miller, "Determining What Individual SUS Scores Mean: Adding an Adjective Rating Scale," *Journal of Usability Studies*, vol. 4, no. 3, pp. 114–123, 2009.
- [35] J. Nielsen and R. Molich, "Heuristic Evaluation of User Interfaces," in *Proc. ACM CHI '90*, 1990, pp. 249–256.
- [36] I. Ajzen, "The theory of planned behavior," *Organizational Behavior and Human Decision Processes*, vol. 50, no. 2, pp. 179–211, 1991.
- [37] R. P. Steel and N. K. Ovalle, "A review and meta-analysis of research on the relationship between behavioral intentions and employee turnover," *Journal of Applied Psychology*, vol. 69, no. 4, pp. 673–686, 1984.
- [38] P. M. Podsakoff, S. B. MacKenzie, J.-Y. Lee, and N. P. Podsakoff, "Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies," *Journal of Applied Psychology*, vol. 88, no. 5, pp. 879–903, 2003.
- [39] S. Barocas and A. D. Selbst, "Big data's disparate impact," *California Law Review*, vol. 104, pp. 671–732, 2016.
- [40] M. Raghavan, S. Barocas, J. Kleinberg, and K. Levy, "Mitigating bias in algorithmic hiring: Evaluating claims and practices," in *Proc. Conf. Fairness, Accountability, and Transparency (FAT\)**, Barcelona, Spain, 2020, pp. 469–481.
- [41] J. Kleinberg, S. Mullainathan, and M. Raghavan, "Inherent Trade-Offs in the Fair Determination of Risk Scores," arXiv:1609.05807, 2016.
- [42] A. Chouldechova, "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments," *Big Data*, vol. 5, no. 2, pp. 153–163, 2017.
- [43] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design Science in Information Systems Research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004.
- [44] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A Design Science Research Methodology for Information Systems Research," *Journal of Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007.

- [45] G. E. A. P. A. Batista, R. C. Prati, and M. C. Monard, "A Study of the Behavior of Several Methods for Balancing Machine Learning Training Data," *ACM SIGKDD Explorations Newsletter*, vol. 6, no. 1, pp. 20–29, 2004.
- [46] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.
- [47] J. Nielsen, "Heuristic Evaluation," in *Usability Inspection Methods*, J. Nielsen and R. L. Mack, Eds. New York: John Wiley & Sons, 1994, pp. 25–62.

Appendices

Appendix A — Research Instrument: The Eight Psychometric Constructs

The local model is fitted on eight constructs derived from the Sri Lankan survey instrument [14]. Each construct is computed as the **arithmetic mean of its constituent Likert items**, each item scored 1–5. The same definition is applied in the research pipeline (`scripts/preprocess_raw.py`) and in the deployed product's Pulse Check scoring, which is what makes train/serve parity possible (§4.7). Table A.1 sets out the composition of each construct.

Table A.1 — Construct composition in the source instrument

Code	Construct	Items	Role
JS	Job Satisfaction	9	Predictor
WLB	Work-Life Balance	5	Predictor
H	Happiness	5	Predictor
MS	Management Support	5	Predictor
CM	Career Management	4	Predictor
IWB	Innovative Work Behaviour	4	Predictor
LMX	Leader-Member Exchange	4	Predictor
CWS	Co-Worker Support	4	Predictor
ET	Employee Turnover Intention	4	Outcome

The outcome variable is the ET composite. `Attrition_binary` is defined as $ET \geq 3.5$ on the 1–5 scale, yielding 33 positive cases in 230 records (14.3%). This cut-point is a design choice, and §5.8 reports model behaviour across alternative cut-points rather than relying on it.

Deployment note. The Pulse Check implemented in the product uses **two items per construct**, not the full battery above. This shortening was a deliberate trade — a 16-item weekly form is answerable, a 40-item one is not — and its consequence is disclosed wherever the Pulse Check is discussed: the short form is a noisier estimate of each construct than the instrument it derives from, and its internal consistency is unassessed because no real responses have been collected (§6.3).

Appendix B — Usability Inspection Evidence Index

Every finding in Table 5.6 is anchored to a captured screenshot. The full evidence set is held in reports/usability/ and indexed below. Two supporting documents accompany it: instrumentation_report.md (the instrumented session log, including response latencies and click paths) and findings_E2_ai_assisted.md (the candidate finding set proposed by evaluator E2 before verification by E1). Table B.1 indexes each screenshot against the findings it supports.

Table B.1 — Screenshot evidence index

File	Screen	Supports
01_login_failed_cons_no_visible_error.png	Login	F2
02_dashboard.png	Dashboard	F12
03_employees_list_no_risk_column.png	Employees	F3, F5
04_attrition_card_idle.png	Attrition risk card	F4, F6, F7
05_attrition_card_scoring_state.png	Attrition risk card	F8
06_attrition_card_result.png	Attrition risk card	F1
07_payroll.png	Payroll	F9, F11
08_login_invalid_credentials_toast.png	Login	F2 (control)
09_dashboard_full_no_risk_element.png	Dashboard	F3
10_pulse_page_clipped_disagree_label.png	Pulse Check	F10
11_payroll_button_count_label.png	Payroll	Verified positive
12_payroll_idempotent_skip.png	Payroll	Verified positive

Image 08 is recorded as a control rather than as a defect: it establishes that a genuinely invalid credential *does* produce a visible error, which is what makes F2 a misattribution finding rather than a missing-feedback finding.

Appendix C — Source Code, Data and Artefact Availability

The datasets used at each stage are listed in Table C.1.

Table C.1 — Datasets

File	Records	Role
data/nexus_hr_master_dataset.csv	2,820	Training (Saudi real 1,191 + Russian real 1,129 + synthetic 500)
data/validation_srilanka.csv	230	Held-out Sri Lankan validation; never used in model fitting, but see the calibration disclosure in §3.5
data/benchmark_ibm.csv	1,470	Published comparison only
data/calibration_params.json	—	Logistic coefficients fitted on 2,550 real records — the 2,320 international records and the 230 validation records (§3.5)

Pipeline. The data pipeline is five Python scripts run in order — `download_datasets.py`, `preprocess_raw.py`, `calibrate.py`, `generate_synthetic_data.py`, `merge_and_clean_data.py` — followed by `train_model.py`, which produces both the transfer and local models together with the SHAP outputs reproduced as Figures 5.1 to 5.5. Each evaluation question in Chapter 5 has its own script and emits a machine-readable JSON report; the figures in this dissertation are generated by those scripts rather than drawn by hand.

Inference service. `ml_service/` is a FastAPI application deployed to Cloud Run with authenticated access only, serving `/predict/local` and `/predict/transfer`. Its preprocessing

mirrors `train_model.py` exactly, and its `requirements.txt` pins the scikit-learn version used in training, because a version mismatch silently breaks model deserialisation.

Product integration. The HR platform consuming these predictions is maintained in a separate repository and integrates over the HTTP API boundary rather than by code sharing. The attrition proxy, the risk card and the Pulse Check described in §4.6 and §4.7 were delivered as pull requests against that repository and are live on its development deployment.

Reproducibility caveat. Trained model bundles and generated report figures are excluded from version control by design; they are reproducible from the scripts and the datasets above. The two real international datasets are redistributed under their original licences and were obtained from their published sources rather than mirrored here.